Journal of Economic Behavior and Organization 164 (2019) 439–468

Contents lists available at ScienceDirect

Journal of Economic Behavior and Organization

journal homepage: www.elsevier.com/locate/jebo

The effects of political competition on the generosity of public-sector pension plans☆

Sutirtha Bagchi

Department of Economics, Villanova University, Villanova, PA, USA

ARTICLE INFO

Article history:
Received 19 July 2018
Revised 29 April 2019
Accepted 18 June 2019

JEL classification:
H75
J45

Keywords:
Public-sector pensions
Political competition
Generosity of benefits
Defined benefit pensions
Defined contribution pensions

ABSTRACT

In politically competitive jurisdictions, there can be strong electoral incentives to increase the generosity of public pensions and simultaneously, to not fund them fully, in order to keep taxes low. I examine the relationship between political competition and generosity of public pensions using a panel dataset for 3000 local plans from Pennsylvania for the period 2003–2013. I find that as the level of political competition in a municipality increases, pension plans become more generous but this relationship holds true only for plans run by municipal governments. A one standard deviation increase in the level of political competition is associated with an increase in the generosity of municipal plans by about 3 percent ($340–504/ retiree/ year) with no effect on plans run by municipal authorities. The effects of political competition are driven by municipalities that have a higher proportion of uninformed voters and are absent for defined contribution plans.

We all know what to do, but we don' t know how to get reelected once we have done it.
-Jean-Claude Juncker, Prime Minister of Luxembourg (2005)[1]

A recent survey of the compensation of state and local government workers (Gittleman and Pierce, 2012) presents us with a number of stylized facts. Although wages for state and local government workers are roughly equal to wages of comparable private-sector workers, benefits are significantly more expensive in the public-sector. These differences in benefits lead to overall compensation costs being 10–19 percent higher in local government (and 3–10 percent higher in state government) than in the private-sector. Honing in on the different categories of benefits, we see that state and local governments spend more than triple on retirement and savings compared to what is spent by employers in the private-sector.[2] Those differences

☆ I thank Sharon Bogden of the Department of General Services of Pennsylvania, Bernard Kozlowski of the Pennsylvania Public Employee Retirement Commission, Rhonda R. Newton of the Pennsylvania Heritage Foundation, Marc Meredith, and the Alliance for Audited Media for help in obtaining data and Daniel Halek for research assistance. I also thank Sourav Bhattacharya, Tilman Borgers, Spencer T. Brien, Charles C. Brown, Joshua Darr, Raymond Fisman, Kyle Handley, James R. Hines Jr., Jeffrey L. Hoopes, Lawrence W. Kenny, Erasmus K. Kersting, Christopher Kilby, Scott Masten, Dana M. Muir, Leslie E. Papke, Giacomo Ponzetto, Stephen Salant, Nathan Seegert, Dan Silverman, Jagadeesh Sivadasan, Joel Slemrod, James M. Snyder Jr., Ugo Troiano, Rodrigo Wagner, and seminar participants at Drexel University, Haverford College, Swarthmore College, the University of Michigan, Towson University, West Virginia University, the International Institute of Public Finance Annual Congress, and National Tax Association Conference for helpful comments. Data acquisition for the project was made possible in part by grants from the Office of Tax Policy Research and the Doctoral Studies Office at the University of Michigan. All links were active as of 04/28/2019. Additional results referenced in the paper but not included here for brevity and copies of the computer programs used to generate the results are available from the author.

E-mail address: sutirtha.bagchi@villanova.edu

[1] http://www.economist.com/node/8808044.

[2] On average, state and local governments spend $3.18 and $3.37/ hour/employee on retirement and savings while the comparable number for the private-sector is $1.00 (Gittleman and Pierce, 2012, Ref. Table 2).

https://doi.org/10.1016/j.jebo.2019.06.014
0167-2681/

are magnified when we examine the amount employers spend on defined benefit (DB) pension plans.[3] It is an open question why compensation for state and local government workers is thus structured – disproportionately in the form of benefits.

This paper offers an explanation that centers on the role of incentives faced by politicians who decide on the terms and conditions of employment for state and local public-sector workers. It proposes that political competition can help explain why public-sector compensation tends to be back-loaded. If public-sector workers are better informed than workers in the private-sector, competition for votes creates incentives for all politicians to promise generous retirement benefits to workers in the public sector and simultaneously, to not fund them fully, in order to avoid raising taxes on workers in the private sector. Put differently, in a competitive environment where there are few partisan voters who vote for a given party regardless of the economic platforms of these parties, politicians attempt to get every vote possible. Consequently they make promises of generous pension benefits to public-sector workers while lacking the political will to raise taxes high enough to cover the full costs of these benefits. As political competition intensifies, these incentives become sharper, resulting in the prediction that a higher degree of political competition is associated with an increase in the generosity of retirement benefits *and* a decline in the funding status of pension plans run by those jurisdictions. For purposes of tractability, this paper examines the effects of political competition on one of those two dimensions – the generosity of pension plans – with the effects of political competition on the funding of such plans the focus of a complementary paper (Bagchi, 2017).

To test the effects of political competition on generosity, I examine municipal pension plans from the state of Pennsylvania. Pennsylvania provides a rich setting to investigate these issues because its local governments offer over 1400 retirement systems that account for more than 40 percent of all public-employee retirement systems nationally.[4] The existence of such a large number of plans is driven in part by the plethora of local governments and by the fact that unlike other states with a large number of local governments, there has never been a consolidation of local plans at the state level. Given the political landscape of the state, we also observe wide variation in the level of political competition, our independent variable of interest. Finally, I am able to obtain and use a high-quality administrative panel dataset spanning the period from 2003 through 2013 for my analysis. These data do not suffer from non-response bias and cover the universe of local pension plans. Given the challenges researchers face in obtaining data on municipal pensions, to the best of my knowledge this is the first paper to analyze the reasons for variation in the generosity of local pension plans.

The primary empirical approach used in the paper is ordinary least squares by which I examine the effect of time-varying measures of political competition at the municipal level on the generosity of DB pension plans run by municipal governments. While I include county and year fixed effects in the baseline specifications, I show that the positive relationship between political competition and pension generosity holds up with alternative empirical strategies as well, either when I introduce municipality and year fixed effects or when I use a simple cross-sectional approach and include each municipal plan only once in the estimation. Those results suggest that there is useful variation in both dimensions – the cross-sectional and the temporal – that is being exploited in our baseline estimates and that our estimates are robust to alternative empirical approaches which make use of only one of these two underlying sources of variation. Our results are robust to including a host of time-varying plan-level controls (such as the actuarial funded ratio and the number of active members) and municipal-level controls (such as per capita income and the level of income inequality). I also observe positive effects of political competition on pension generosity when I use alternate measures of generosity or when I measure political competition differently. Lastly, these results are robust to the inclusion of county-by-year fixed effects or municipality-specific linear time trends.

Beyond presenting the effects of political competition on the generosity of municipal pension plans, I examine the effects of political competition on DB plans that are run by municipal authorities. In contrast to the governing body of a municipality that is directly impacted by the outcome of an election and is likely to be responsive to voters, municipal authorities are headed by *appointed* boards which are relatively far removed from the electoral will of voters. The insulation of authorities from political influence along with their ability to establish pension plans that are subject to the identical reporting and funding standards as municipalities enables me to use municipal authorities as a placebo. In contrast to the positive relationship between political competition and pension generosity observed for municipal plans, there is no relationship between the level of political competition and generosity for plans run by municipal authorities. In terms of magnitude, a one standard deviation increase in the level of political competition is associated with an increase in the generosity of municipal plans by about 3 percent (or $340–504 per retiree per year) with *no* effect on plans run by municipal authorities.[5]

An added attraction of using data on local pensions from Pennsylvania is that about a quarter of the plans are defined contribution (DC).[6] Given that a more generous DC plan requires higher employer contributions in the present – which must be met through tax increases likely to displease voters – we expect the effects of political competition on such plans to be

[3] DB plans promise lifetime pension benefits that are typically a specific fraction of an employee' s last drawn salary. State and local governments spend $2.65 and $3.09/ hour/ employee on DB plans whereas the private-sector spends only $0.43 (Gittleman and Pierce, 2012, Ref. Table 2).

[4] https://www.census.gov/prod/2013pubs/g11-aspp-sl.pdf (Ref. Table 5a). Of the 3418 public-employee retirement systems in the United States, 1422 (or, 41.6%) are local retirement systems from Pennsylvania.

[5] A one standard deviation increase in the level of political competition, using the measure defined in Besley et al. (2010), would result if the Democratic vote share were to go down from 57.5 percent (leaning Democratic) to 50 percent (most competitive), or go up from 42.5 percent (leaning Republican) to 50 percent (most competitive).

[6] "Defined contribution plans are retirement plans that specify the level of employer contributions, if any, and place those contributions in individual accounts. The value of an individual account is determined by the amount of money contributed and the rate of return on the money invested over time" (BLS, 2010)

muted compared to the effects of political competition on DB plans. Those predictions are confirmed in the data. I find that political competition has *no* effect on the employer contribution rate for DC plans, and this holds for DC plans offered by municipalities as well as DC plans offered by municipal authorities.

Finally, I examine an assumption made in the construction of models that associate the underfunding of public pensions with a lack of information and understanding among private-sector voters about public pensions (Glaeser and Ponzetto, 2014). I propose that back-loading DB pension plans (and underfunding them) can persist in equilibrium only if private-sector workers are not fully informed of the benefits promised to public-sector workers and do not realize that they will have to bear the burden of any shortfalls in the pension fund through a combination of tax increases and service cuts. I take this implication seriously and split the sample of municipalities into two groups based on the level of voter awareness and engagement with the prediction that the effects of political competition are smaller in places that have more well-informed voters. Using data on the prevalence of newspaper readership at the local level as a proxy for voter awareness, I find that the effects of political competition are indeed muted (and in fact, absent) in municipalities where a larger fraction of residents subscribe to a newspaper. The results from this sample-split test suggest that when faced with an electorate that is more informed of the true cost of offering back-loaded compensation packages to public-sector workers, politicians are less likely to make these pension plans more generous.

1. Literature review

This paper builds on a number of different streams of literature. In the body of work that examines public-sector labor markets, much of the focus has been on differences in wages between public- and private-sector workers (see, for example, Ehrenberg and Schwarz, 1986 and the studies cited therein). However, given the growing importance of benefits for all workers, particularly for those in the public sector, examining wages alone is unlikely to provide a complete picture of the differences in compensation between the two sectors. More recently, Gittleman and Pierce (2012) and Bewerunge and Rosen (2013) have used microdata to investigate the how wage and pension benefits compare for workers in the public- and the private-sectors. The papers arrive at roughly similar conclusions even though they look at different periods and use very different data sources.[7] Both report that while wages of state and local workers are roughly equal to that of comparable private-sector workers, benefits are significantly more generous in the public-sector and drive overall compensation costs higher for that sector.[8]

Focusing most directly on the question of why compensation structures are back-loaded, explanations have been proposed that back-loading may promote employee retention and foster longer on-the-job tenures, which in turn enhances productivity (Lazear, 1990; Lazear and Moore, 1988). The empirical evidence in support of that thesis is however mixed. Gustman and Steinmeier (1993) note that jobs covered by pensions also offer higher levels of compensation than what workers can obtain elsewhere, and it is these compensation premiums (rather than the existence of pensions per se) that drive lower turnover. Moreover, turnover rates are lower for jobs offering DC plans as well which tend to not be backloaded. Gustman and Steinmeier (1995) hint that an unmeasured factor associated with employment on jobs offering backloaded pensions may account for some of the reduction in turnover observed for workers covered by pensions. The implication for us is that as we compare the generosity of pension benefits across municipalities, we control for a large set of employer and employee attributes.

Moving further afield, the basic argument offered in this paper that the lack of understanding about pensions creates incentives for politicians to offer generous benefits ties to the idea of fiscal illusion. Under fiscal illusion taxpayers' failure to perceive the full extent of tax burdens can prevent them from recognizing the true cost of public services and as a result, distort fiscal policy (e.g. Banzhaf and Oates, 2013; Congleton, 2001).[9] Political budget cycles by which politicians run expansionary monetary and/or fiscal policies prior to elections can be seen as a manifestation of fiscal illusion. While the first generation of models (e.g., Nordhaus, 1975) rested on some form of irrationality or ignorance on the part of voters, subsequent models consider voters as fully rational but imperfectly informed (e.g. Rogoff, 1990; Rogoff and Sibert, 1988). The takeaway from this body of literature is simply that politicians pander to voters by running hidden deficits and the more responsive voters are to pandering, the higher the hidden deficits. In the context of Pennsylvania, where local governments are constrained to run balanced budgets, politicians enact policies that lead to deficits in the pension plans they run.

Tied to the literature on the political economy of budget deficits, a number of papers have examined the influence of politics on the fiscal health of public pensions.[10] Fitzpatrick (2017), for example, exploits a reform enacted by the Illinois

[7] Gittleman and Pierce (2012) use the 2009 Current Population Survey (CPS) and the 2009 Employer Costs for Employee Compensation (ECEC) microdata collected as part of the National Compensation Survey (NCS) while Bewerunge and Rosen (2013) use data from the 2004 and 2006 waves of the Health and Retirement Study (HRS) and focus on workers 50 and older.

[8] Bewerunge and Rosen (2013) include federal workers in their analysis as well and find that wages of federal workers are about 28 log points higher than comparable private-sector workers. These higher wages are *not* offset by lower retirement benefits and federal employees have substantially more pension wealth than their private-sector counterparts.

[9] Fiscal illusion has its origins in the writings of Puviani (1903) but received significant interest following Buchanan (1960, 1967), and is reviewed in Oates (1988).

[10] Although I do not include in our present discussion, a related and relevant literature examines the politics of Social Security. See for example, Congleton and Shughart (1990) and Mulligan and Sala-i Martin (1999).

state legislature in 2005 that required school districts to bear the full cost of increasing wages for teachers approaching retirement to find support for the view that intergovernmental incentives distort the true costs of offering generous retirement benefits to public-sector workers. Focusing squarely on the politics of pensions, Anzia and Moe (2017) note that prior to the Great Recession voters were unconcerned and uninformed about public pensions creating an environment "conducive to a bipartisan brand of politics in which Republicans went along with Democrats and unions in supporting generous pension plans for public workers, which they drastically underfunded."[11] Most relevant to us in providing a theoretical underpinning for this study is Glaeser and Ponzetto (2014) who posit that pension obligations are shrouded because of lower availability of information about pensions than wages and because of the greater difficulty taxpayers face in understanding the operation of defined benefit plans, in contrast to current compensation.[12] While a complete discussion of their model is beyond the scope of this paper, their basic message applies to our context as well: given voters' lack of understanding of DB pensions, the political process induces the back-loading of compensation in the form of retirement benefits.[13]

2. Institutional context

Before turning to the empirical analysis, it is worthwhile having a brief primer on the institutional context. As noted earlier, I examine local pension plans from the state of Pennsylvania because it provides a rich setting; the state offers more than thrice the number of public-sector retirement systems as any other state and accounts for two-fifths of the nation' s public-sector plans. The existence of such a large number of plans in Pennsylvania can be attributed to its complex system of local government. With over 2500 municipalities – cities, boroughs, and townships – the state has the second highest number of general purpose local governments in the country.[14] Moreover, unlike other states with a large number of municipalities, most local governments in Pennsylvania establish separate pension plans for their police and non-uniformed employees and are not part of a larger state-wide plan.[15] The advantages of using municipal data to test the hypotheses are the large number of comparable cases that share the same national and state-level political context (e.g. state income tax rates) at the same time they exhibit wide variation on the variables of interest, viz. political competition and pension generosity. The availability of rich municipal-level data from the Decennial Censuses and the American Community Surveys (ACS) also enables me to control for many potentially important municipal characteristics that could influence the generosity of pension plans.

Another aspect of Pennsylvania's municipal pensions that makes it attractive to analyze is that municipal authorities, that number over 1500 (Department of Community and Economic Development, 2015), can and do offer pension plans subject to the same reporting and funding standards as municipalities. Authorities tend to perform a very limited number of functions such as the provision of water (e.g. Erie City Water Authority) or sanitation services (e.g. Radnor–Haverford–Marple Sewer Authority). Municipalities justify the provision of services through a municipal authority on multiple grounds. Beyond the single-minded focus that the board of an authority can have on its operations and the fact that many services are provided efficiently only if a large service area spanning multiple municipalities is covered, the governance of municipal authorities may also be more conducive to their operation.[16] Authority board members are appointed by the governing body of the municipalities where they provide service for five-year overlapping terms. Therefore in contrast to the governing body of a municipality that can change *en masse* following an election, the appointed board of an authority is relatively insulated from the electoral will of voters. As a result, Bennett and Dilorenzo (1982) note that authorities "can raise and spend money without reference to the immediate wishes of the electorate, whereas a government can raise and spend money only in the amounts and manner specified by the electorate under the constitution and statutes of the state." The insulation of municipal authorities from political influence along with their ability to establish pension plans for their employees that are governed by similar statutes enables me to use municipal authorities as a "control group" and contrast the effects of political competition on municipal plans from the effects on plans run by municipal authorities.

Examining the effects of political competition for municipal authorities requires assigning each authority to a single municipality. Doing so is straight-forward for an authority that serves a single municipality where the service area of an authority and a municipal boundary perfectly match (e.g. Pittsburgh Public Parking Authority). In the case of municipal authorities that service more than one municipality (e.g. the East Norriton Plymouth Joint Sewer Authority), the following

[11] Kelley (2014) also examines the underlying causes for pension underfunding of state-level public pension plans and finds that a combined model which encompasses the median voter theorem, along with the theory of "capture" by special interest groups, provides the strongest explanation for the current levels of unfunded liabilities, with the special interest group model slightly outperforming the median voter model in head-to-head comparisons.

[12] In support of their claim, they note that salaries of state employees are publicly disclosed annually whereas no such database exists for the accruing pensions of retirees or active members. For example, salary data for state employees from the state of Pennsylvania are available at: http://www.pennlive.com/midstate/index.ssf/2013/03/search_pennsylvania_state_empl.html.

[13] A similar intuition is provided in Epple and Schipper (1981) who conjecture that increased political competition may induce politicians to underfund pension liabilities, so as to be able to reduce taxes in the short-run; this behavior is rewarded by those voters who are unaware of deferred pension obligations.

[14] Source: https://www.census.gov/content/dam/Census/library/publications/2012/econ/g07-cg-isd.pdf, p. 252.

[15] Larger municipalities may also have a separate pension plan for firefighters. Teachers belong to a separate state-wide system, the Pennsylvania Public School Employees Retirement System (PSERS) that is not a part of this analysis.

[16] For example, of the 213 municipal authorities which offer Defined Benefit (DB) plans and appear in 2013, the last year of our sample, about 70.0% are sanitary authorities or water authorities or authorities that provide both services simultaneously and these authorities "commonly serve more than one municipality to take advantages of the economies of scale and natural drainage areas." (Department of Community and Economic Development, 2015).

rules were used in order to assign a municipal authority to a single municipality. First, I examine the composition of the board of the authority and assign the authority to that municipality which has the largest number of board seats. If that information is unavailable or is inconclusive (e.g. equal number of board members from two or more municipalities), I examine how the authority is financed; I assign the authority to whichever municipality (or municipal residents) pays the largest share of expenses associated with the operation of the authority – information that is obtained from the annual reports of these authorities. If such information is not available either or is yet inconclusive, then I assign the authority to that municipality which has the largest population among all the municipalities served by the authority. Full details of the assignment are available on request from the author.[17]

3. Empirical analysis

3.1. Construction of variables and data sources

Data regarding municipal pension plans offered by the various local governments within Pennsylvania were obtained from the Public Employee Retirement Commission (PERC). As part of its enabling statute, the PERC was required to publish a status report on a biennial basis that lists the level of assets and liabilities and the number of active members for each local government plan.[18] While these reports are available to us from 1985 onwards,[19] they lack the data necessary for analyzing the generosity of pension benefits – for example, the normal costs of the plans or the amount paid out in benefits to retirees. The dataset that includes those variables, although only available from 2003 through 2013, are rich and are what I use for my analysis. They include the name of the municipal entity offering the plan, the type of plan (DB vs. DC), the employee group covered (either policemen or firefighters or non-uniformed personnel), and the status of coverage by Social Security.[20] They also include the number of active members and the corresponding payroll, the number of beneficiaries and the amount paid out in benefits, and the amount contributed by employees into the plan. The dataset for DB plans includes the funded ratio and the normal costs for the plan along with the interest rate assumed in estimating actuarial liabilities while the dataset for DC plans includes the contributions made by employers into the plan – both in absolute terms and as a percentage of payroll.

Constructing measures of political competition at the local level is challenging as there is no central repository for data on municipal elections at either the federal or the state level. I construct proxy measures for political competition at the local level by looking at the vote shares for the two parties for all races to any of the six offices for which elections are held on a state-wide basis, namely, U.S. President, U.S. Senator, Governor,[21] Attorney General, Auditor General, and Treasurer.[22] Data on votes cast for each of these offices for candidates from both the Republican and Democratic parties (and any other parties that may have contested) are available at the level of each individual municipality in successive issues of the Pennsylvania Manual. Because the results for a particular candidate in any one election cycle may have a large idiosyncratic component to it, I average the Democratic vote share[23] across all elections held within a given year to any of the six offices in constructing the average Democratic vote share for that year. For example, in constructing a measure of political competition for a municipality for 2003, I examine all state-wide races held in 2002 to any of the six offices for that municipality. Following Besley et al. (2010), the key measure of political competition I use in the paper is defined as: $PC_{mt} = -|D_{mt} - 0.5|$ where m indexes municipality, t indexes year, and D_{mt} is the average Democratic vote share in municipality m in year t.

Before proceeding further, it is worth examining the reasonableness of using data on national and state elections to construct measures of political competition at the local level. We can ascertain how justifiable that is by examining the correlations between the limited data available for outcomes of local races and races to national and state-level offices for the same time period. Using a dataset on the composition of municipal councils, I find a correlation coefficient of 0.6525 ($p < 0.001$) between the share of council seats held by Democrats in 2009 and the average Democratic vote share for all national and state races held in 2008. Bagchi (2017) offers additional evidence of a positive and statistically significant relationship between the average Democratic vote share in municipal elections and the average Democratic vote share for

[17] The null results for authorities are robust to the quality of the underlying match. Tables A.1 and A.2 in the Appendix present splits for DB and DC plans based on whether a municipal authority matches with a municipality unambiguously or not.

[18] The PERC was created through the Public Employee Retirement Commission Act of 1981 to "review legislation affecting public employee pension and retirement plans and to study on a continuing basis public employee pension and retirement policy as implemented at both the State and local level, the interrelationships of the several systems and their actuarial soundness and cost" (http://www.legis.state.pa.us/WU01/LI/LI/US/PDF/1981/0/0066..PDF). The responsibilities of the commission were transferred to the Department of the Auditor General in 2016 (http://www.legis.state.pa.us/cfdocs/legis/li/uconsCheck.cfm?yr=2016&sessInd=0act=100).

[19] Bagchi (2017) which analyzes the effects of political competition on the funding level of pension plans uses these data and finds that higher levels of political competition are associated with declines in the funded ratio for municipal plans.

[20] About 28 percent of state and local government employees in the U.S. were not covered by Social Security in 2008 (Nuschler et al., 2011). I find that extent of coverage of local employees within Pennsylvania under Social Security is similar to the national average, with about 26 percent of local employees in the sample not covered by Social Security in 2009.

[21] Election for the office of Lieutenant Governor is held separately in the primary election; for the general election each party' s ticket for Governor and Lt. Governor is made up of the highest vote getters in the separate primary elections.

[22] As Besley et al. (2010) note, name recognition of candidates for down-ballot offices is typically very low among voters, making it likely that measures of political competition based on races for these offices is driven largely by party attachment of voters rather than the popularity of individual politicians.

[23] Defined as Votes cast for Democrats/ (Votes cast for Democrats + Votes cast for Republicans).

Table 1
Summary statistics.

Variable	Units	Mean	Median	Standard deviation	Minimum	Maximum
Pension plan characteristics						
For DB municipal plans:						
Average annual pension	In dollars	16,111	14,341	9988	1692	38,544
Normal costs	As a percent of payroll	12.53	12.19	4.71	3.76	23.05
Ratio of benefits to wages	As a fraction	0.32	0.31	0.16	0.022	1.38
For DB plans run by municipal authorities:						
Average annual pension	In dollars	12,355	9958	8254	1692	38,544
Normal costs	As a percent of payroll	9.92	9.44	4.02	3.76	23.05
Ratio of benefits to wages	As a fraction	0.26	0.23	0.15	0.025	1.08
For DC plans run by municipalities:						
Employer contribution rate	As a percent of payroll	7.93	7.72	3.30	2.19	15.50
For DC plans run by municipal authorities:						
Employer contribution rate	As a percent of payroll	7.74	7.00	3.45	2.19	15.50
Plan-level controls						
Coverage in Social Security	0 = No. 1 = Yes	0.771	1	0.420	0	1
Employees covered by collective bargaining	In percent terms	34.35	33.33	32.68	0	88.24
Controls at the municipal level						
Per capita income	In dollars	25,583	23,996	7931	14,416	46,643
Taxes spent on debt servicing	In percent terms	19.19	11.60	22.20	0.00	93.12
Unemployment rate	In percent terms	6.99	6.35	2.98	2.08	14.65
Households that are owner-occupied	In percent terms	69.51	70.31	14.35	44.43	94.07
Population aged 65 or older	In percent terms	16.92	16.85	4.23	8.60	28.31
Revenues per capita	In dollars	837	724	490	145	2135
Income inequality	In absolute terms	1.26	1.25	0.10	1.06	1.50
Ethnic fragmentation	In absolute terms	1588	1129	1336	0	4753
Political variables						
Democratic vote share	As a fraction	0.477	0.467	0.133	0.139	0.892
Political Competition	As defined in text	−0.112	−0.103	0.075	−0.392	−0.000

Summary statistics for the dependent variables, average benefits per retiree, normal costs, and ratio of benefits to wages are based on biennial data from 2003 to 2013 provided by the Pennsylvania PERC. Data on Social Security coverage also come from the Pennsylvania PERC. The percentage of employees organized under collective bargaining is for 1982 from the Employment Summary Statistics of Census of Governments. Per capita income, unemployment rate, percentage of households that are owner-occupied, percentage of the population aged 65 or older, income inequality, and ethnic fragmentation are from the Census and the ACS as described in footnote 24. Following Alesina et al. (2000), income inequality is defined as the ratio of mean household income to median household income and ethnic fragmentation is defined as 10,000 − sum of squares of percentages of the population from each race. Taxes spent on debt servicing and revenues per capita are based on annual data from 2003 to 2013 from the Pennsylvania DCED. All of these variables have been winsorized at the 2.5% and 97.5% levels. Lastly, the political variables, average Democratic vote share and measures of political competition are based on all elections to national and state-level offices held in even-numbered years and are constructed using successive issues of the Pennsylvania Manual. The precise election years used depend on the lag structure employed, as described in the text.

elections held to national and state-level offices. All in all, examining national and state races seems to offer an accurate picture of local politics in Pennsylvania.

A number of additional data sources are used for our analysis. Data on controls at the municipal level that might affect pension plan generosity, such as the per capita income of the area, are drawn from the 2000 Decennial Census, the 2007–2011 and the 2011–2015 American Community Survey (ACS) 5-year estimates.[24] I also include the share of tax revenues spent on debt servicing as a proxy of municipal fiscal health and revenues per capita as a proxy for government size. Those variables are constructed using municipal financial reports prepared on an annual basis by the Pennsylvania Department of Community and Economic Development. The fraction of employees covered by collective bargaining is based on the 1982 Employment Summary Statistics of Census of Governments. The lack of more recent data on unionization, while not ideal, is not very concerning because while union membership rose rapidly in the 1960s and 1970s, by the early 1980s it had stabilized in an equilibrium that still prevails (Anzia and Moe, 2015).[25]

Summary statistics for all variables are presented in Table 1 below. The table indicates the large amount of variation in the dependent variables of interest: the annual average pension benefit, the normal cost, and the ratio of benefits to

[24] The 2000 Census provides data on municipal demographic controls for the year 2000. The 2007–2011 5-year ACS, being centered on 2009, is used for that year, while the 2011–2015 ACS is used for 2013. I use a linear interpolation for other years. The 5-year estimates from the ACS are used because they provide data for all areas, whereas the 1-year (3-year) estimate only covers areas with populations more than 65,000 (20,000). https://www.census.gov/programs-surveys/acs/guidance/estimates.html To put that in context, the median-sized municipality in Pennsylvania has fewer than 2000 residents.

[25] Pennsylvania has allowed collective bargaining for policemen and firefighters since 1968 and for other employees since 1970. (http://www.dli.pa.gov/INDIVIDUALS/LABOR-MANAGEMENT-RELATIONS/plrb/Pages/default.aspx).

wages for DB plans and the average employer contribution rate for DC plans. I also note the considerable variation in the level of political competition observed from −0.392 (corresponding to a Democratic vote share of 0.892 in Yeadon Borough, Delaware County for 2001 – least competitive) to −0.000 (corresponding to a Democratic vote share of 0.500 in Roaring Brook Township, Lackawanna County for 1993 – most competitive).[26] An observation one can draw from this table is that while DB plans offered by municipalities tend to be more backloaded than similar plans offered by municipal authorities – they have higher normal costs and a higher ratio of benefits to wages on average – the same does not appear to be true for DC plans. Municipal DC plans appear similar to DC plans offered by municipal authorities on the basis of the employer contribution rate to such plans.

3.2. Empirical specification

Panel data on plan features for all municipal DB and DC plans are available on a biennial basis over the period from 2003 to 2013. The primary empirical approach used is ordinary least squares in which I include all observations corresponding to DB plans run by municipalities and estimate the effects of political competition on the generosity of these municipal DB plans. I then estimate a separate set of regressions for all DB plans run by municipal authorities and offer those results as a contrast to the effects of political competition for municipal plans. The empirical specification is:

$$G_{imt} = \alpha + \beta_1 * PC_{m(t-l)} + \beta_2 * W_{it} + \beta_3 * E_{it} + \beta_4 * C_{it} + \beta_5 * X_{it} + \beta_6 * Z_{mt} + \lambda_c + \gamma_t + \varepsilon_{imt} \qquad (1)$$

where:

- G_{imt} is the dependent variable, for example, the average annual benefit per retiree for plan i in municipality m (or a municipal authority serving municipality m) in year t;
- PC_{mt} is a measure of political competition in the municipality m for year t, with l the lag length;
- W_{it} and E_{it} are the log of wages and the employee's contribution to the pension plan as a percentage of payroll;
- C_{it} are a set of dummy variables indicating which group of employees are covered by the plan (e.g. policemen or non-uniformed personnel, etc.);
- X_{it} includes the coverage of employees under Social Security (1 = Yes, 0 = No), the fraction of employees covered by collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries;
- Z_{mt} are time-variant controls at the municipal level listed below;
- λ_c are county fixed effects and γ_t are year fixed effects; and
- ε_{imt} is the error term.

In the empirical analysis, I start off with a parsimonious specification that includes the measure of political competition, county and year fixed effects, employee-group dummies, and log of wages and member contributions to the pension plan. I control for the log of wages to account for the possibility that more generous pension benefits may be offset by lower wages. Second, unlike DB plans in the private-sector, employees covered by DB plans in the public-sector contribute to these plans (Munnell et al., 2007). Thus, a pension plan could be more generous in the sense that it pays out more in retirement benefits but that may simply reflect higher contributions made by the employee during her years of service, rather than the tendency of politicians to pander and make public pensions more generous as proposed. Accordingly, I control for the employee contribution rate to the pension plan.

A richer specification includes controls for the coverage of employees under Social Security because plans where participants are not covered by Social Security tend to be more generous (Munnell et al., 2007). This richer specification also controls for the fraction of employees covered by collective bargaining given the evidence that points to a positive association between unionization and fringe benefits (see, for example, Lewis, 1990 and the studies cited therein), although recently questions have been raised about whether the positive relationship is causal or whether it is driven by unobserved heterogeneity across unionized and non-unionized workforces (e.g. Frandsen, 2016; Lovenheim, 2009).[27] I also control for the actuarial funded ratio of the plan as it seems plausible that in jurisdictions that have better funded pension plans politicians will have an incentive (and the ability) to make these plans even more generous. Lastly, I control for the log of number of active members and beneficiaries as politicians may be more keen to raise benefit levels when public-sector employees constitute a larger interest group as they tend to vote in local elections at rates significantly higher than the general public (e.g. Johnson and Libecap, 1991).[28]

[26] Using the 11-year lag structure described in the following section, pension data for the year 2013 are matched with measures of political competition based on elections held in 2001. Likewise, elections held in 1993 are matched with pension generosity as of 2005.

[27] My interest is in measuring the causal impact of political competition on benefits and so I do not take a stance on whether the observed union premiums reflect the causal effects of unions on benefits or whether unobserved differences across municipalities drive those differences. By including coverage under collective bargaining as a control I simply want to reduce the possibility that differences across plans based on unionization are driving the effects of political competition that we observe.

[28] A back-of-the-envelope calculation (assuming full turnout by active members and beneficiaries of public pension plans) suggests that municipal employees and retirees may have cast between 17.0 and 34.5 percent of votes in the 2011 Mayor elections in Philadelphia and the 2013 Mayor elections in Pittsburgh. While the proportion of votes cast by public-sector workers is typically smaller in less populous municipalities, in close elections, for ex-

The most complete set of specifications include time-varying controls at the municipal level that might affect the generosity of pension plans – the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation. I control for the percentage of households that are owner-occupied because owners have a longer time horizon than renters and we may therefore expect homeowners to exercise more discipline on politicians who make decisions about public-sector benefits (Fischel, 2001; Oliver and Shang, 2007). Per capita income, the local unemployment rate, and the fraction of tax revenues spent on debt service are included as controls as proxies for local economic conditions; prosperous municipalities are likely to offer generous pensions whereas municipalities experiencing high levels of fiscal stress may find it harder to do so. I include the percentage of population aged 65 or older as a control for the age structure of the population as municipalities with a larger fraction of older voters may be more willing to simply pass on these obligations to future generations. The log of revenues per capita is included as a control as to address the possibility that the municipalities that are offering more generous benefits are simply those municipalities that tend to spend more in general. Lastly, the level of income inequality and ethnic fragmentation are included as controls because there is some evidence that politicians use public-sector employment as a redistributive device in cities that are more unequal and that have a higher level of ethnic fragmentation (Alesina et al., 2000). I also include dummy variables for the class to which a municipality belongs (for example, Township versus Borough versus City) as municipalities of different classes vary in the set-up of their local governments and that might also influence pension plan characteristics.

The question of how to lag political competition in the empirical specification is not one that admits an easy answer. The data on pension benefits are not disaggregated by retiree; we simply know the number of retirees and what they receive in benefits. This group is likely to include retirees from various cohorts: individuals who retired within the last year as well as those who retired 25 years earlier and are still drawing pension benefits. Decisions about what set of benefits each cohort of retirees would receive were decided at various points in time during the municipality's history. Complicating matters further, decisions about benefit increases are often made retroactively as was the case for state-government workers in California in 1999[29] and in Pennsylvania in 2001.[30] Accordingly it becomes challenging for us to associate the average level of benefits received by retirees with the level of political competition at a particular point in time that in turn corresponds to a unique lag structure.

I therefore look to the literature for guidance. The most recent work in the area (Rauh, 2017) suggests that the effective average duration of pension liabilities is 11 years whereas prior work suggested an average duration of 13 years (Novy-Marx and Rauh, 2011) and 15 years (Waring, 2004a; 2004b). Thus, in our baseline specifications I lag political competition by 11 years but in the robustness checks, I present results with several different lag structures ranging from 0 to 4 years on the lower end and 13 or 15 years on the upper end which implicitly make different assumptions about the pace of adjustment. Fortunately, our results are not particularly sensitive to the lag structure used.

Given the possibility of inter-temporal correlation in the error terms, I cluster standard errors at the county level all throughout (Bertrand et al., 2004). Doing so tends to typically yield the most conservative standard errors – standard errors that are larger than those obtained by clustering either at the municipal level or at the plan level or along two dimensions – by county and year or by municipality and year.

4. Examining the effects of political competition on defined benefit plans

In this section, I present estimates of the effects of political competition on the generosity of DB pensions operated by the various municipalities and municipal authorities within Pennsylvania. I start off with results from a parsimonious specification, which is then augmented with the inclusion of variables that are likely to influence the generosity of pension plans. I next present a host of robustness checks which examine the sensitivity of the results to operationalizing the dependent variable differently or to measuring political competition in alternative ways. I conclude this section with a number of robustness checks that address possible concerns of omitted variable bias.

Results from the base specification on pension plan generosity

Before providing the regression results, I first present the data visually. In Fig. 1 I simply plot the average benefit levels for municipal plans and plans run by municipal authorities splitting the sample into terciles[31] based on their underlying level of political competition. Given that benefit levels differ substantially between non-uniformed personnel versus policemen and firefighters, I plot benefit levels for these different groups of employees separately, with retiree benefits for non-uniformed personnel in the two panels on the left and those for policemen and firefighters in the two panels on the right.

ample, the 2013 Mayor election in Phoenixville Borough where only 29 votes separated the two candidates, public-sector workers can often swing the outcome of an election. Thus, even in terms of sheer numbers, public-sector workers constitute an important interest group. Additionally, as Sieg and Wang (2013) document in an empirical analysis of municipal elections in the 150 largest cities in the U.S. between 1990 and 2012, challengers that receive union endorsements strongly benefit from them in competitive races via a higher likelihood of being elected.

[29] https://www.city-journal.org/html/pension-fund-ate-california-13528.html.

[30] http://triblive.com/news/editorspicks/8530945-74/pension-state-billion & https://www.pasr.org/pa-house-pension-reform/.

[31] Terciles partition the data into three equal-sized groups, each containing a third of the total data.

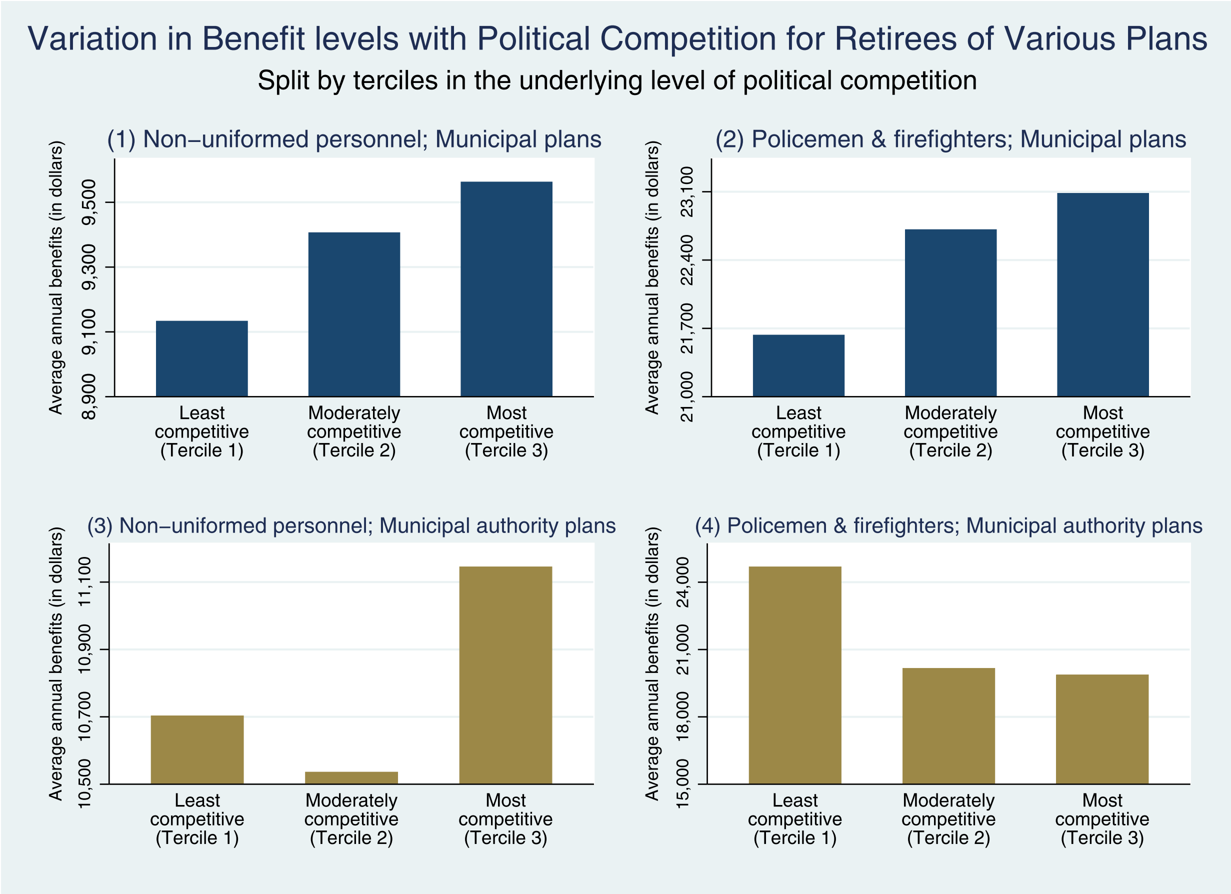

Fig. 1. Variation in benefit levels with political competition for municipal plans and plans run by municipal authorities for various employee-groups. All plans included in the estimation in Table 2 were included for this analysis and then split into three equal-sized groups (or "terciles") based on their underlying level of political competition. For each of the four panels, I plot the mean benefit levels for plans that correspond to particular combinations of entity and employee-groups and fall in that tercile.

An inspection of the figure conveys the key message of this paper: We observe that an increase in political competition is associated with an increase in the generosity of benefits for municipal pension plans for both groups of employees, but there is no consistent pattern in the relationship between political competition and the level of benefits for plans operated by municipal authorities. The rest of this section is devoted to showing that these patterns in the data hold up to the inclusion of a variety of controls which might affect plan generosity and are robust to several plausible empirical specifications.

I present the regression results which are obtained by estimating specification (1) with the log average annual pension benefit per retiree as the dependent variable in Table 2. Columns (1) through (3) of the table present the coefficient on political competition for municipalities while columns (4) through (6) present the coefficient for municipal authorities. Columns (1) and (4) correspond to the most parsimonious specification that includes the measure of political competition, county and year fixed effects, employee-group dummies, and log of wages and member contributions to the pension plan. Columns (2) and (5) introduce a dummy variable for the coverage of employees under Social Security, the fraction of employees covered by collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Columns (3) and (6) are the most complete specification and include the class of the municipality and all municipal-level controls that proxy for local economic conditions, the size of government, homeownership, the age structure of the population, local income inequality, and ethnic fragmentation. Collectively, these results form our baseline specifications.

The coefficients on political competition in columns (1) through (3) suggest that a higher level of political competition is associated with a higher level of benefits for retirees in municipal plans. To provide a sense of magnitude of these effects, a one standard deviation increase in the intensity of political competition[32] is associated with an increase in the benefit

[32] A one standard deviation increase in the level of political competition, using the measure defined in BPS (2010), would result if the Democratic vote share were to go down from 57.5 percent (leaning Democratic) to 50 percent (most competitive), or conversely, if the Democratic vote share were to go up from 42.5 percent (leaning Republican) to 50 percent (most competitive).

Table 2
Effect of political competition on log of average benefits (Baseline specifications).

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Political Competition	0.418***	0.306**	0.282**	0.136	0.139	0.184
	(0.135)	(0.140)	(0.137)	(0.215)	(0.249)	(0.303)
Log of wages	1.012***	0.692***	0.618***	1.089***	0.911***	1.005***
	(0.0853)	(0.103)	(0.100)	(0.205)	(0.212)	(0.228)
Member contributions to the plan	0.0569***	0.0382***	0.0408***	0.0182	0.0250*	0.0287*
	(0.00686)	(0.00510)	(0.00513)	(0.0145)	(0.0148)	(0.0158)
Social Security coverage		0.00633	−0.00226		−0.103	−0.145
(0 = No coverage, 1 = Coverage)		(0.0293)	(0.0315)		(0.0994)	(0.0973)
Fraction of employees covered under		0.111**	0.0962**		0.0616	0.0134
collective bargaining		(0.0423)	(0.0391)		(0.146)	(0.153)
Log of number of active members		0.0927***	0.0813**		0.145**	0.129
		(0.0334)	(0.0336)		(0.0718)	(0.0779)
Log of number of beneficiaries		0.0248	0.0435		−0.00660	−0.00931
		(0.0267)	(0.0297)		(0.0693)	(0.0664)
Actuarial funded ratio		−0.00116**	−0.00116**		0.000729	0.000318
		(0.000511)	(0.000518)		(0.00137)	(0.00139)
Percent housing units owner-occupied			−0.00300*			−0.00743
			(0.00162)			(0.00543)
Log of per capita income			0.192**			0.233
			(0.0944)			(0.246)
Percentage of population aged 65 or older			0.00533			0.00809
			(0.00338)			(0.0110)
Unemployment rate			−0.00317			0.0258
			(0.00621)			(0.0158)
Fraction of tax revenues spent on debt service			0.0798**			−0.0197
			(0.0348)			(0.116)
Log of revenues per capita			−0.0126			0.0120
			(0.0218)			(0.0950)
Level of income inequality			−0.0637			−0.179
			(0.159)			(0.445)
Level of ethnic fragmentation			0.0160			−0.0296
			(0.0122)			(0.0438)
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Number of observations	8359	8359	8359	1141	1141	1141
R^2	0.54	0.57	0.57	0.47	0.49	0.51

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. County and year fixed effects, along with employee-group dummies are included in all specifications. Dummy variables for the class of municipality are included in columns (3) and (6). The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

received by the typical retiree for a typical pension plan by about 2.1–3.1 percent. Given average annual benefits of $16,111, this translates to an increase of about $340–504.[33] In contrast, based on examining the coefficients in columns (4) through (6), we do not find any effect of political competition on the generosity of plans run by municipal authorities, the estimated coefficients being statistically indistinguishable from zero.[34] The results from Table 2 can be captured in the form of Fig. 2, which illustrates that political competition has a positive effect on the generosity of municipal plans, but not on plans run by authorities.

The coefficients on some of the control variables, while not of primary interest, are worth mentioning. Consistent with much of the empirical literature, the coefficient on wages in these benefit regressions is positive and statistically significant across all specifications. Pension benefits are also more generous if employees contribute a greater fraction of their wages into the DB plan with a one percent increase in employee contributions associated with a five percent increase in benefits. Moving from no coverage under collective bargaining to full coverage of all employees is associated with an increase in pension benefits by 9.6 percent in our most complete specification. Economic prosperity of the municipality, as captured by

[33] E.g. 0.418 * 0.0749 = 0.0313. Given average benefits of $16,111 for municipal plans, this translates to $504/ retiree/ year.

[34] The standard errors are however large enough that I cannot rule out the possibility that the coefficients on political competition for municipal plans are equal to the corresponding coefficients for municipal authority plans.

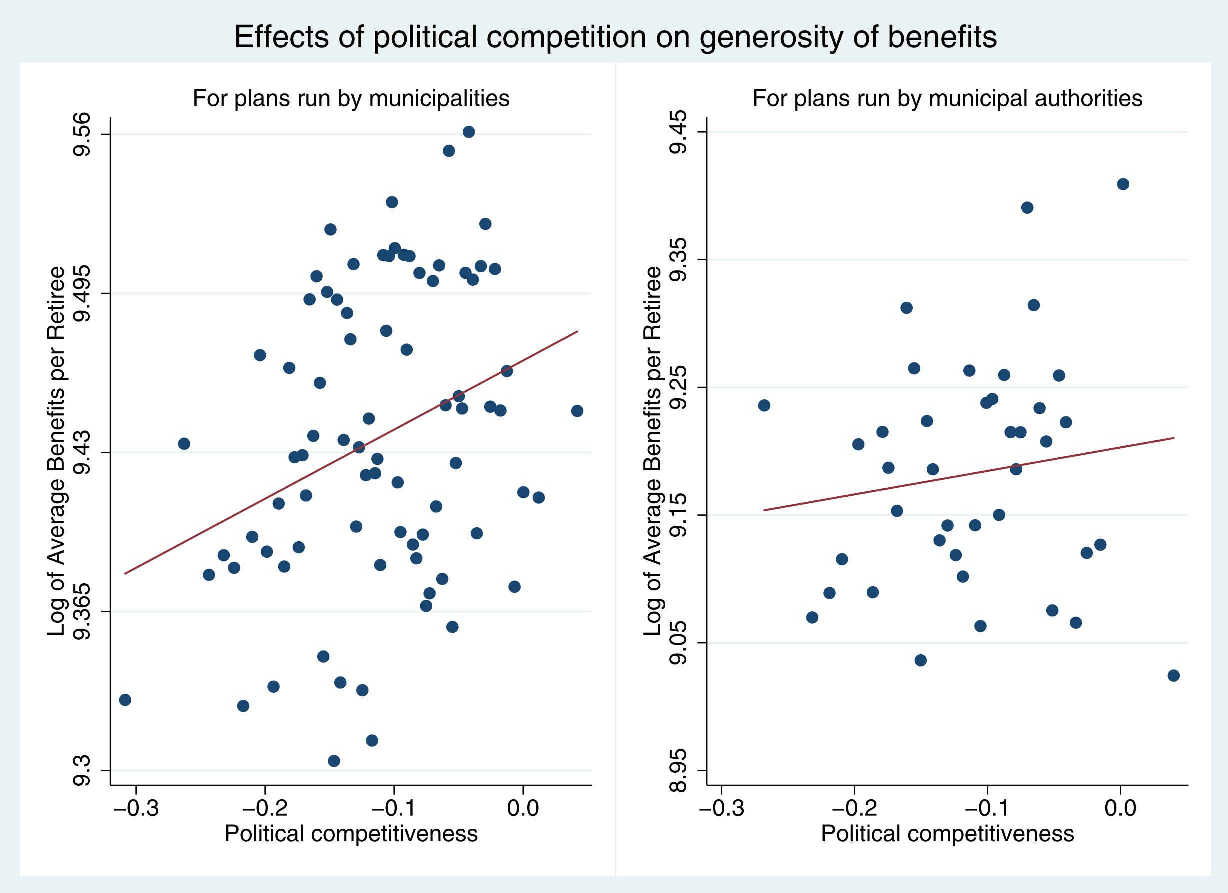

Fig. 2. Effects of political competition on benefit levels for municipal plans and plans run by municipal authorities. These plots were drawn using the Stata command, binscatter, with 80 bins used for municipal plans and 40 bins used for municipal authority plans. They correspond to the regression results presented in Columns (3) and (6) of Table 2 in which I regress log average benefits on political competition, controlling for log wages, member contributions to the plan, employee-group dummies, plan-level controls (a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries) and socioeconomic controls (the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation), along with dummy variables for the class of municipality, and county and year fixed effects.

the per capita income, is associated with a more generous pension plan,[35] while a higher prevalence of homeownership has a moderating influence on the generosity of these benefits.[36]

Two variables which show up as statistically significant and whose signs differ from our *a priori* expectations are the actuarial funded ratio and the debt service variables. I find a negative relationship between the funded ratio and the generosity of municipal pension plans. This may be because the incentives that induce politicians to make pension plans more generous similarly induce them to underfund these plans. I also find that municipalities where a greater fraction of the tax base goes towards paying interest on the debt have *more,* rather than less generous pension benefits perhaps reflecting the fact that municipalities with generous pensions end up borrowing more.[37]

A possible concern about these results is that municipalities with a pre-existing culture of fiscal restraint may have set up authorities in order to insulate certain functions from electoral competition. In that case, there is an omitted variable bias; it is this culture of fiscal restraint that dampens the generosity of pension benefits for municipal authorities, rather than the mere fact that a municipal authority exists. To address this potential concern, I examine the legislative history and the reasons why municipal authorities were set up in Pennsylvania. I find that municipal authorities in the state had their beginning during the Depression of the1930′s when the federal government granted money to states and municipalities

[35] A one standard deviation increase in per capita income is associated with an increase in benefits by about 6 percent.

[36] A one standard deviation increase in the fraction of municipal residents who are homeowners is associated with a decrease in the generosity of benefits by about 4 percent.

[37] Revenues per capita, included as a proxy for government size, is however statistically insignificant.

for public works construction to stimulate employment and provide needed public facilities. Pennsylvania was one of only three states to pass general enabling legislation in 1935 that allowed its municipalities to create authorities (Department of Community and Economic Development, 2015). Although there have been numerous changes over the years to the statutory framework including the passage of the present Municipality Authorities Act in 1945, the history suggests that many municipal authorities were created at a time far removed from the period I analyze.[38]

I also attempt to address the concern about omitted variable bias econometrically. First, I note that the null results for municipal authorities are robust to the inclusion of municipality fixed effects. Second, under the reasonable assumption that municipalities with a pre-existing fiscal culture of restraint are likely to have a smaller government, I include a control for the log of revenues per capita as a proxy for government size. Doing so does not affect either the statistical or economic significance of our results and the variable is statistically insignificant in nearly all specifications. Lastly, to address this issue of why authorities emerge, I estimate a series of regressions that examine factors which might correlate with the propensity of a municipality to establish an authority. Based on these regressions, I do not find any evidence that suggests politically competitive municipalities set up authorities to insulate themselves from electoral competition. The only variable which comes across as statistically significant across a range of specifications is municipal size; larger municipalities appear more likely to establish a municipal authority. That result is intuitive; municipalities incur fixed costs in setting up authorities and given that, it only makes sense for the larger municipalities to incur these fixed costs in establishing municipal authorities. Collectively, these results address the endogeneity concern that municipalities set up municipal authorities to insulate themselves from electoral competition.

Understanding the source of identifying variation

Our baseline specification includes multiple observations for each municipal plan and regresses the average benefits paid out by these plans against time-varying measures of political competition. An important question is about the source of the identifying variation, i.e. does the variation in political competition being exploited in generating these estimates come from variation *across* municipalities or variation in the intensity of political competition *for a given municipality over time*? To address that question, I present Table 3 which includes the baseline results with county and year fixed effects and then experiments with two alternative empirical approaches. In columns (4) through (6), I present results obtained by introducing municipality and year fixed effects whereas in columns (7) through (9), I use a simple cross-sectional approach in which each plan is included only once.

Based on the observation that the coefficients on political competition in the first row of the table continue to be statistically significant regardless of the empirical approach used, we can conclude that: (1) An increase in political competition *for a given municipality* is associated with an increase in the generosity of these benefits but also that (2) Comparing *across municipalities*, more politically competitive municipalities have more generous pensions. Thus, there is useful variation in both dimensions – the cross-sectional and the temporal – that is being exploited in our estimates and our results are robust to alternative empirical strategies which make use of only one of these two sources of variation.

4.1. Robustness checks

In this subsection, I examine the robustness of the finding that political competition is associated with an increase in the generosity of DB municipal plans but not for plans run by municipal authorities.

4.1.1. Alternative measures of generosity of DB plans

The primary measure of the generosity of pension plans I use in the analysis is the average annual benefit per retiree in that plan. The reason I prefer it over other measures, such as the normal cost which measures the present value of benefits accrued in a given year, is because such measures can often be systematically and significantly lower than they ought to be.[39] Sabin (2015) in a study of public plans from California reports that the main cause of underfunding in pension plans for the city of Sunnyvale – the municipality he analyzes – was underestimating the normal rate of contribution. As per his analysis, over the 18-year period from fiscal years 1995 to 2012, the normal rate prescribed by CalPERS should have been set 9.0% higher as a percentage of payroll in order to fully fund the plans. It was the understatement of normal costs, rather than the poor performance of retirement assets, which accounted for about three-quarters of the unfunded liability for Sunnyvale over this period.[40] Additionally, plan sponsors use different assumptions in arriving at an estimate of plan liabilities, with some plans assuming an interest rate of 5.5 percent, and others assuming an interest rate of 8 percent.

These limitations of the normal cost notwithstanding, I examine the effects of political competition on the normal costs of local pension plans from Pennsylvania as this measure has been used in prior work (e.g. Munnell et al., 2011). In Panel A of Table 4 I use the numbers as-is, that is, as reported by the government entities themselves, which implicitly assumes that

[38] Even a 1964 report of the Advisory Commission on Intergovernmental Relations shows Pennsylvania with the third highest number of special authorities – next only to Illinois and California. https://library.unt.edu/gpo/acir/Reports/policy/A-22.pdf.

[39] The normal cost is calculated by apportioning the total present value of an employee's expected benefits in retirement to each year of an employee's worklife, based on a specific actuarial cost method (Public Plans Data, 2001–2016).

[40] Exhibit 8 of Sabin (2015) shows that for Sunnyvale, investment loss (=$35.3M) + normal loss (= $217.3M) + amortization loss (= $40.8M) = unfunded liability ($293.4M). The normal loss thus accounts for 74.1 percent of the unfunded liability.

Table 3
Effect of political competition on log of average benefits using alternative empirical strategies.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Baseline specification			Using municipal fixed effects			Using a cross-section		
Political Competition	0.418***	0.306**	0.282**	0.149**	0.191***	0.168**	0.608**	0.482*	0.480*
	(0.135)	(0.140)	(0.137)	(0.0660)	(0.0673)	(0.0668)	(0.276)	(0.266)	(0.270)
Log of wages	1.012***	0.692***	0.618***	0.423***	0.305**	0.303**	1.156***	0.862***	0.851***
	(0.0853)	(0.103)	(0.100)	(0.123)	(0.137)	(0.138)	(0.0952)	(0.112)	(0.114)
Member contributions	0.0569***	0.0382***	0.0408***	0.0451***	0.0370***	0.0370***	0.0602***	0.0432***	0.0445***
to the plan	(0.00686)	(0.00510)	(0.00513)	(0.00851)	(0.00719)	(0.00724)	(0.00841)	(0.00702)	(0.00728)
Social Security coverage		0.00633	−0.00226		−0.0399	−0.0403		0.00879	0.0138
(0 = No cov., 1 = Coverage)		(0.0293)	(0.0315)		(0.0537)	(0.0538)		(0.0379)	(0.0391)
Fraction of employees covered		0.111**	0.0962**		Drops out as measure			0.110**	0.0829*
under collective bargaining		(0.0423)	(0.0391)		is time-invariant			(0.0495)	(0.0490)
Log of number of		0.0927***	0.0813**		0.0254	0.0259		0.0780**	0.0773**
active members		(0.0334)	(0.0336)		(0.0378)	(0.0380)		(0.0334)	(0.0364)
Log of number of		0.0248	0.0435		0.140***	0.142***		−0.00192	−0.00691
beneficiaries		(0.0267)	(0.0297)		(0.0409)	(0.0409)		(0.0227)	(0.0285)
Actuarial funded ratio		−0.00116**	−0.00116**		−0.00152**	−0.00152**		−0.00120**	−0.00124***
		(0.000511)	(0.000518)		(0.000624)	(0.000623)		(0.000458)	(0.000462)
Percent housing units			−0.00300*			0.00253			−0.00129
owner-occupied			(0.00162)			(0.00246)			(0.00190)
Log of per capita income			0.192**			0.0197			0.0786
			(0.0944)			(0.114)			(0.0977)
Percentage of population			0.00533			−0.00759*			0.00610
aged 65 or older			(0.00338)			(0.00419)			(0.00414)
Unemployment rate			−0.00317			−0.00504			−0.00324
			(0.00621)			(0.00351)			(0.00873)
Fraction of tax revenues			0.0798**			0.00678			0.176**
spent on debt service			(0.0348)			(0.0216)			(0.0774)
Log of revenues per capita			−0.0126			0.0210			−0.0476
			(0.0218)			(0.0224)			(0.0314)
Level of income inequality			−0.0637			−0.0175			0.136
			(0.159)			(0.119)			(0.191)
Level of ethnic fragmentation			0.0160			0.00776			0.0168
			(0.0122)			(0.0204)			(0.0122)
Fixed effects introduced	County and Year Fixed Effects			Municipality and Year Fixed Effects			County Fixed Effects		
Employee-group dummies	✔	✔	✔	✔	✔	✔	✔	✔	✔
Number of observations	8359	8359	8359	8359	8359	8359	1579	1579	1579
R^2	0.54	0.57	0.57	0.57	0.58	0.58	0.56	0.58	0.58

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. Employee-group dummies are included in all specifications. Dummy variables for the class of municipality are included in columns (3), (6), and (9). The fraction of employees organized under collective bargaining is sourced from the 1982 Employment Summary Statistics of Census of Governments and being time-invariant, it drops out when I introduce municipal fixed effects. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

Table 4
Effect of political competition on normal costs.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Normal costs unadjusted (as-is)						
Political Competition	1.962*	2.012*	2.033**	0.0874	0.204	−0.430
	(1.003)	(1.013)	(0.979)	(1.868)	(1.619)	(1.474)
Number of observations	10,697	10,697	10,697	1521	1521	1521
R^2	0.41	0.42	0.43	0.55	0.59	0.60
Panel B: Normal costs recalibrated assuming a 7 percent interest rate and duration of liabilities = 11 years						
Political Competition	2.515***	2.313**	2.261**	−0.101	−0.0599	−0.600
	(0.929)	(0.943)	(0.941)	(1.798)	(1.794)	(1.586)
Number of observations	10,697	10,697	10,697	1521	1521	1521
R^2	0.47	0.48	0.49	0.59	0.61	0.63
Panel C: Normal costs recalibrated assuming a 7 percent interest rate and duration of liabilities = 13 years						
Political Competition	2.613***	2.366**	2.303**	−0.136	−0.107	−0.631
	(0.929)	(0.937)	(0.940)	(1.804)	(1.837)	(1.617)
Number of observations	10,697	10,697	10,697	1521	1521	1521
R^2	0.47	0.49	0.50	0.60	0.62	0.63
Panel D: Normal costs recalibrated assuming a 7 percent interest rate and duration of liabilities = 15 years						
Political Competition	2.711***	2.419**	2.345**	−0.170	−0.155	−0.660
	(0.933)	(0.933)	(0.942)	(1.816)	(1.884)	(1.650)
Number of observations	10,697	10,697	10,697	1521	1521	1521
R^2	0.48	0.50	0.50	0.60	0.62	0.63
Log of wages	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable in Panel A is the normal costs reported by the plan "as-is". Subsequent panels adjust the reported normal costs with respect to an interest rate of 7 percent, corresponding to the median across all municipal pension plans in the sample following the steps laid out in footnote [41]. The weighted average duration of liabilities assumed in the recalibration exercise are 11 years, 13 years, and 15 years in Panels B, C, and D respectively. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. County and year fixed effects, employee-group dummies, and log of wages are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

the projections used by municipalities in coming up with estimates of normal costs are uncorrelated with their underlying levels of political competition. Diebold et al. (2018) suggests that this may not be the case; the authors note that when normal costs increase, "administrators tend to use less prudent methods that defer, or keep low, the pension contributions required from the state while, simultaneously, and perversely, improving the appearance of the plan's funded status and the state's funding discipline."

Attempt to recalculate the normal costs of a pension plan on the basis of a common interest rate follow Novy-Marx and Rauh (2011). On the basis of those steps, I recalculate the normal costs with respect to an interest rate of 7 percent, corresponding to the median across all municipal pension plans in the sample. Panels B, C, and D of Table 4 present the results for three choices of the weighted average duration of liabilities – 11 years, 13 years, and 15 years.[41] Columns (1) through (3) provide the coefficient on political competition for municipalities while the coefficient for municipal authorities is provided in columns (4) through (6).

As we can see, the coefficients on political competition continue to be positive and statistically significant in each of the panels for municipal plans. A one standard deviation increase in the intensity of political competition is associated with an

[41] Here is an example: If the normal costs for a plan assuming an 8 percent interest rate are 12 percent (of payroll), the normal costs recalibrated to 7 percent equal 0.12 * (1.08/1.07)∧11, or 13.3 percent, *if the average duration of liabilities is 11 years*. If the average duration of liabilities is longer at 13 years, the normal costs are recalculated as 0.12 * (1.08/1.07)∧13 = 13.5 percent.

Table 5
Effect of political competition on ratio of benefits to wages, log of ratio of benefits, and normal costs adjusted for variations in interest rate assumptions and employee contributions to the plan.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Dependent variable: Ratio of benefits to wages						
Political Competition	0.0841**	0.0671*	0.0692*	0.0119	0.0194	0.0354
	(0.0389)	(0.0377)	(0.0372)	(0.0598)	(0.0617)	(0.0701)
Number of observations	8359	8359	8359	1141	1141	1141
R^2	0.30	0.32	0.32	0.30	0.31	0.32
Panel B: Dependent variable: Log of Ratio of benefits to wages						
Political Competition	0.419***	0.289**	0.273**	0.138	0.140	0.184
	(0.136)	(0.131)	(0.129)	(0.217)	(0.245)	(0.302)
Number of observations	8359	8359	8359	1141	1141	1141
R^2	0.29	0.32	0.33	0.29	0.33	0.35
Panel C: Dependent variable: Normal costs adjusted for variations in interest rate assumptions and member contributions to the plan						
Political Competition	4.913**	3.998*	4.078*	0.262	0.155	−1.370
	(2.164)	(2.028)	(2.084)	(4.238)	(4.264)	(3.821)
Number of observations	10,697	10,697	10,697	1521	1521	1521
R^2	0.48	0.52	0.52	0.53	0.56	0.58
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries.		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable in Panel A is the ratio of the average benefits received by retirees to the wages received by active members, while in Panel B, the dependent variable is the log of that ratio. In Panel C, the dependent variable is a proxy for the value of pension benefits over a retiree's lifetime calculated as described in the text. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. County and year fixed effects and employee-group dummies are included in all specifications. Employee contributions to the pension plan (as a percentage of payroll) are included in Panels A and B. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

increase in the normal costs of about 0.2 percent or 1.2 percent of the average normal costs. In contrast, there are no effects of political competition on the normal costs for pension plans run by municipal authorities, with the coefficients negative throughout in Panels B through D although they are far from statistical significance.[42]

Beyond looking at the average benefits received by a retiree and the normal costs of the plan, another way of capturing the degree to which DB plans are back-loaded is to examine the ratio of benefits to wages. Constructing such a ratio is fraught with similar issues we encounter earlier when contemplating the ideal lag structure; the benefit levels pertain to retirees whereas wages pertain to members who are currently active and working and data limitations prevent us from obtaining wages for retirees or benefit levels for those who will be retiring in the future. With that important caveat in mind, I construct two measures – the ratio of benefits to wages and the log of that ratio – and examine the effects of political competition on these dependent variables in Panels A and B of Table 5.

I also construct another measure of benefit generosity which tries to capture the value of these pension benefits over a retiree's lifetime.[43] This measure is defined as:

$$(Total\ Normal\ Cost * [(1 + expected\ return) \wedge duration)/(1 + risk\ free\ interest\ rate) \wedge duration] \\ - Member\ Normal\ Cost \tag{2}$$

where the total normal cost includes the employer's and employee's contribution to the plan expressed as a percentage of payroll and the expected return is the pension-specific assumed interest rate. In this expression, the first term is the

[42] As the 7 percent rate is likely too high, I also discount them back to an interest rate corresponding to the nominal yield on zero-coupon Treasury bonds of similar duration. Based on recent market conditions and expectations of market participants, I use 1.5 percent for the real yield on long-term zero-coupon Treasury bonds and add in 2 percent to reflect inflation expectations, for a nominal yield of 3.5 percent. I find similar results when calibrating normal costs at a 3.5 percent interest rate.
[43] I am grateful to an anonymous referee for this suggestion.

total future value of expected benefits to be paid over a worker's retirement, discounted by a risk-free rate to reflect the virtually guaranteed nature of these benefits.[44] The workers' contributions are then netted out to captures differences in the contributions workers are making to their own retirement.

Based on the coefficients on political competition in Table 5, we note that political competition is associated with an increase in the ratio of benefits to wages and the measure of lifetime value of pension benefits for municipal plans. This statement however does not apply for plans run by municipal authorities and in fact, in Panel C, we note that the coefficient on political competition for plans run by municipal authorities changes signs.

4.1.2. Alternative ways of measuring political competition

Beyond using the measure of political competition based on Besley et al. (2010), one can operationalize political competition differently. As Boyne (1998) points out, one ought to take the volatility of party strength into account when constructing a measure of political competition. Therefore, a potential alternative measure is the standard deviation of Democratic vote share across all elections that occur in any given year within a given municipality. Results with political competition, thus defined, as the independent variable are presented in Panel A of Table 6.

As we can see, political competition continues to have a positive and statistically significant effect on the average benefit received per retiree. A one standard deviation increase in the level of political competition, using this measure, leads to an increase in the average retirement benefit of about 3 percent for municipal plans, with no effect for pension plans run by municipal authorities.

The second approach I use in terms of operationalizing political competition differently is to introduce a linear and squared term for the average Democratic vote share in the same specification. If political competition makes pension plans more generous, then one would expect to see a positive coefficient on the linear term and a negative coefficient on the squared term generating an inverted U-shape. This is, in fact, what I find with this alternative operationalization of political competition in Panel B. Based on my estimates, the Democratic vote share which corresponds to the highest level of benefits varies in a range from 49.5 to 58.7 percent, with the 95% confidence intervals including 50 percent in all specifications. Thus, benefits increase as the Democratic vote share increases to about 55 percent and then they decline.[45]

The third approach I use in terms of operationalizing political competition differently involves redefining the Democratic vote share by factoring in votes received by third party-candidates as well. Thus, I redefine the average Democratic vote share as Votes cast for Democrats/ (Votes cast for Democrats + Votes cast for Republicans + Votes cast for third-party candidates) and recompute our measure of political competition. This does not make a large difference to either the average Democratic vote share or the measure of political competition[46] and therefore, not surprisingly, when I introduce this measure of political competition in the regressions, my estimates are similar to those reported earlier. A one-standard deviation increase in political competition is associated with a 3 percent increase in the benefits received by retirees of municipal plans, with no effect for plans run by municipal authorities.

Thus the results in Table 6 confirm the positive relationship between political competition and benefit levels for municipal pensions that we observe in Table 2. They also support the lack of a similar positive relationship between political competition and benefits for plans run by municipal authorities.

To address the question of whether the partisan leaning of a municipality has an independent effect on the relationship between political competition and the generosity of pensions, I look for data on the political affiliation of the Mayor or the President of the Borough Council or the Township Board in order to classify municipalities as "Democratic-majority" or "Republican-majority". Data on political affiliations of Mayors and Council Members are hard to come by for smaller municipalities. However, I am able to obtain the name of the Mayor or the Council President (for municipalities without a mayor) for 2004 using the Pennsylvania Manual and use voter registration data to establish the political affiliation of these elected officials.[47] I use that information to split the sample of municipalities into those which had (a) a Republican Mayor or Council President or (b) a Democratic Mayor or Council President in 2004. I include political affiliation of the Mayor or Council President as an independent variable in Panel A whereas in Panels B and C, I conduct the estimation separately for municipalities with a Republican Mayor (or Council President) and a Democratic Mayor (or Council President) respectively. These results are included in Table 7 and they suggest that it is higher political competition at the municipal level, rather than partisanship, that results in more generous benefits.[48]

[44] The Pennsylvania Supreme Court has applied the state's constitutional ban against the impairment of contracts to protect the rights of public employees to their pensions. To the best of my knowledge, not a single municipality or municipal authority in the state has defaulted on its pension obligations, even when they have been under significant fiscal stress or in receivership.

[45] For example, using the coefficients in col. (2), the level of Democratic vote share at which benefits are maximized = $1.235/(2^{*}1.247) = 49.5$ percent and the p-value that the implied point of maximum equals 50 percent is 0.9442.

[46] The median value of the variable, average Democratic vote share, goes down from 46.7 percent to 43.9 percent.

[47] I am grateful to Marc Meredith for sharing the voter registration data for 2008 that were used in Meredith and Grissom (2010) and that form the primary source used for categorizing these elected officials. Additionally consulting Internet sources as needed, I am able to successfully match 98.8% of the 8359 municipal observations that were used in our baseline specification. Two reasons account for the non-matches: first, there were a few individuals serving in office in 2004 who had passed away or moved and did not show up in the 2008 state voter registration data. Second, there were vacancies for two municipalities: Hatfield Borough and Codorus Township.

[48] The coefficients in Panel B are statistically indistinguishable from corresponding coefficients in Panel C. I can also use the Democratic vote share to split municipalities into Democratic-majority (vote share > 50%) and Republican-majority (vote share <= 50%). The results from such a split are generally similar to those included here and are available on request.

Table 6
Effect of political competition on log of average benefits using alternative measures of political competition.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Measure of political competition: Standard deviation of Democratic vote share						
Coefficient on standard deviation of Dem. vote share	0.870*	0.714*	0.665	0.338	0.472	0.787
	(0.441)	(0.419)	(0.413)	(1.640)	(1.446)	(1.388)
R^2	0.54	0.57	0.57	0.47	0.49	0.51
Panel B: Measure of political competition: Democratic vote share - linear and squared						
Coefficient on Democratic vote share	2.059***	1.235**	1.286***	1.124	0.212	−0.240
	(0.469)	(0.469)	(0.435)	(1.204)	(1.067)	(1.265)
Coefficient on Democratic vote share squared	−1.754***	−1.247**	−1.154**	−0.698	−0.379	−0.366
	(0.459)	(0.481)	(0.469)	(1.017)	(0.944)	(1.079)
Democratic vote share at which benefits are maximized	58.7%	49.5%	55.7%	80.5%	28.0%	−32.8%
95% Confidence Interval for the above Dem. vote share	[47.0, 70.5]	[36.2, 62.9]	[38.3,73.1]	[−9.4, 170.4]	[−127.2, 183.0]	[−555.3, 489.8]
R^2	0.55	0.57	0.57	0.47	0.50	0.51
Panel C: Defining the Democratic vote share by including votes received by third parties & recalculating BPS (2010) measure of political competition						
Political Competition	0.478***	0.285**	0.285**	0.190	0.0467	0.0322
	(0.127)	(0.126)	(0.124)	(0.258)	(0.249)	(0.291)
R^2	0.54	0.57	0.57	0.47	0.49	0.51
Number of observations	8359	8359	8359	1141	1141	1141
Log of wages and employee contributions to pensions	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). The measure of political competition used in Panel A is the standard deviation of Democratic vote share for all elections held to national and state offices within a given year. In Panel B, I use a non-parametric approach including a linear and squared term for the Democratic vote share. In Panel C, I redefine the average Democratic vote share as Votes cast for Democrats/ (Votes cast for Democrats + Votes cast for Republicans + Votes cast for third-party candidates) and recompute the BPS (2010) measure of political competition, viz. $PC_{mt} = -|0.5 - D_{mt}|$. All measures of political competition have been lagged by 11 years. County and year fixed effects, employee-group dummies, log of wages, and employee contributions to the pension plan (as a percentage of payroll) are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

4.1.3. Other specification checks

This sub-section includes a number of checks that address any residual concerns of the robustness of our results. As before, columns (1)–(3) of Table 8 present the estimated effect of political competition on plan benefits for municipalities while columns (4)–(6) present the estimates for municipal authorities.

Robustness Check (RC) 1: Controlling for average Democratic vote share. In the regressions estimated thus far, I have not included the average vote share for Democrats as a control variable. Introducing that however lets us separately identify the effect of an increase in Democratic support from an increase in the level of political competition. Upon doing so, I find that the coefficient on average Democratic vote share is statistically insignificant all throughout (and in fact, changes signs) whereas an increase in political competition continues to be associated with an increase in the benefit levels for municipal plans with no effect on plans run by municipal authorities.

RC2: Using vote shares based solely on Presidential elections. Voters may consider the performance of their local officials in casting their votes for elections to state-level offices (e.g. Governor). This raises the possibility of reverse causality between the fiscal health of a municipal plan and the level of political competition in that municipality. Voters are however unlikely to consider the performance of their local officials as they decide who to vote for President. Thus, using vote share based

Table 7
Effect of political competition on log of average benefits controlling for political affiliation of the Mayor/ Council President.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Controlling for the political affiliation of the Mayor/ Council President in addition to our measure of political competition						
Political Competition	0.407***	0.299**	0.275*	0.111	0.0982	0.142
	(0.136)	(0.143)	(0.141)	(0.221)	(0.250)	(0.296)
Political affiliation of the Mayor/ Council President	0.00705	−0.0235	−0.0120	0.0916	0.0428	0.0337
(0 = Republican, 1 = Democrat)	(0.0382)	(0.0378)	(0.0401)	(0.124)	(0.113)	(0.128)
Number of observations	8261	8261	8261	1127	1127	1127
R^2	0.55	0.57	0.57	0.47	0.50	0.51
Panel B: Limited to municipalities (and corresponding municipal authorities) that have a Republican Mayor/ Council President						
Political Competition	0.718***	0.406**	0.421**	−0.452	−0.696	−0.597
	(0.205)	(0.187)	(0.164)	(0.343)	(0.437)	(0.456)
Number of observations	4877	4877	4877	581	581	581
R^2	0.56	0.59	0.60	0.58	0.61	0.63
Panel C: Limited to municipalities (and corresponding municipal authorities) that have a Democratic Mayor/ Council President						
Political Competition	0.367**	0.441***	0.355*	0.445	0.752	0.528
	(0.144)	(0.162)	(0.209)	(0.475)	(0.515)	(0.472)
Number of observations	3384	3384	3384	546	546	546
R^2	0.58	0.60	0.61	0.50	0.54	0.55
Log of wages and employee contributions to pensions	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). Panel A includes a control for the political affiliation of the Mayor or the Council President. Cities and boroughs in Pennsylvania generally have Mayors, while townships of the first and the second class have boards or councils that are headed a Council President/ Chairperson. There are exceptions as municipalities can adopt a Home Rule Charter or an Optional Plan which permits local governments to adopt a different structure than prescribed by statute. Panel B is limited to municipalities (and corresponding municipal authorities) which have either a Republican Mayor or a Republican Council President, while Panel C is limited to those entities that have either a Democratic Mayor or a Democratic Council President. Data on the political affiliation of the Mayor or the Council President are obtained from the 117th Volume of the Pennsylvania Manual and are as of 2004. All measures of political competition have been lagged by 11 years. County and year fixed effects, employee-group dummies, log of wages, and employee contributions to the pension plan (as a percentage of payroll) are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

solely on Presidential elections minimizes the possibility of reverse causality associated with using data on elections to all national and state-level offices.[49] Using just such a measure of political competition, I find that the coefficients on political competition are similar to their previous values.

RC3: Including county-by-year fixed effects. The baseline specifications presented include county (and year) fixed effects, which allow us to control for time-invariant factors at the county level that might affect the generosity of municipal pension plans within that county. To lend further credibility to our identification strategy, I introduce county-by-year fixed effects that allow for each county to be hit by a unique shock in each year. I find that our results are robust to the inclusion of these county-by-year fixed effects, with the coefficients on political competition somewhat larger in magnitude.

RC4: Including municipality-specific linear time trends. The next check is particularly demanding of the data and is geared to addresses concerns about secular unobserved trends that might simultaneously lead to changes in pension generosity and in political competitiveness of a given municipality. In this robustness check, I introduce a linear time trend for each municipality in our sample thus allowing the level of retirement benefits to evolve separately for each municipality. Results

[49] Using a 11-year lag structure also reduces the possibility of reverse causality.

Table 8
Robustness checks for the effect of political competition on log of average benefits.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Base Specification	0.418***	0.306**	0.282**	0.136	0.139	0.184
	(0.135)	(0.140)	(0.137)	(0.215)	(0.249)	(0.303)
RC1: Controlling for average Democratic vote share	0.411***	0.306**	0.278*	0.161	0.129	0.163
	(0.134)	(0.140)	(0.140)	(0.252)	(0.236)	(0.276)
RC2: Using vote shares based solely on Presidential elections	0.378**	0.420**	0.359*	−0.155	0.195	0.354
	(0.164)	(0.175)	(0.182)	(0.481)	(0.503)	(0.551)
RC3: Introducing county-by-year fixed effects	0.518***	0.374**	0.350**	0.198	0.203	0.264
	(0.172)	(0.170)	(0.168)	(0.367)	(0.425)	(0.532)
RC4: Introducing municipality-specific linear time trends	0.159**	0.170**	0.157**	0.0372	0.00703	−0.0105
	(0.0698)	(0.0688)	(0.0706)	(0.181)	(0.206)	(0.190)
RC5: Long differences (only including data for years 2003 and 2013) with municipality fixed effects	0.306*	0.379**	0.317*	0.295	0.242	0.0764
	(0.166)	(0.165)	(0.186)	(0.624)	(0.641)	(0.629)
RC6: Only including municipalities that have at least one authority	0.580***	0.448**	0.282*	0.136	0.139	0.184
	(0.155)	(0.182)	(0.162)	(0.215)	(0.249)	(0.303)
RC7: Weighted least squares with log of actuarial assets as weights	0.395***	0.292**	0.265*	0.112	0.131	0.175
	(0.130)	(0.141)	(0.137)	(0.199)	(0.240)	(0.296)
Log of wages and employee contributions to pensions	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013, unless otherwise noted. The number of observations equals 8359 for the base specification, RC1, RC2, RC3, RC4, and RC7, 2666 for RC5, and 2941 for RC6 for municipal plans (cols. (1)–(3)). The number of observations equals 1141 for the base specification, RC1, RC2, RC3, RC4, RC6 and RC7, and 375 for RC6 for plans run by municipal authorities (cols. (4)–(6)). The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. County and year fixed effects, employee-group dummies, log of wages, and employee contributions to the pension plan (as a percentage of payroll) are included in all specifications, unless otherwise noted. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

with municipality-specific trends continue to indicate a positive effect of political competition on the generosity of benefits, although the estimated effects are smaller in size. By contrast, the coefficients on political competition for municipal authorities are precisely estimated to be zero and they change signs depending on the controls.

RC5: Long differences. In order to examine if long-run shifts in the intensity of political competition have the same effect as those found using data for all years, I estimate regressions using data from only the first (2003) and the last (2013) year of the sample. I additionally combine the use of long differences with municipality fixed effects. The thought experiment underlying the use of municipality fixed effects is whether an increase in the level of political competition for a given municipality results in an increase in the generosity of benefits for that municipality. Given that political competition for a municipality is likely to change slowly over time, the use of a long differences approach seems sensible. The positive relationship between the intensity of political competition and benefit levels for municipal plans holds with this approach as well, and the coefficients are similar in magnitude to those in the base specifications with no statistically significant effects for pension plans run by municipal authorities.

RC6: Only including municipalities that have at least one authority. This robustness check is motivated by the fact that not all municipalities which offer a pension plan also have an authority offering their own pension plan. In particular, municipalities that have a municipal authority offering their own pension plan are larger than the typical municipality offering a pension plan. Hence this robustness check limits the estimation only to pension plans run by municipalities with at least one municipal authority. Although this causes our sample size to drop to about a third of the original size, we continue to observe that an increase in political competition is associated with an increase in the generosity of municipal pension plans.

RC7: Using weighted least squares. Whether to use weights or not in a regression when our goal is to recover the causal effect of interest is not entirely clear (Solon et al., 2015) who accordingly suggest reporting both weighted and unweighted estimates as best practice. Our regressions thus far have been unweighted, thereby according equal importance to a plan with a single member and a plan with several hundred members. To explore whether the results hold if I assign weights to plans based on their size, I re-estimate the regressions with weights equal to the log of actuarial assets in the plan.[50] The coefficients on political competition for municipal plans are similar to what we had earlier suggesting that the effects of political competition are present across plans of varying sizes. In contrast, there are no effects of political competition on the generosity of pension plans run by municipal authorities.

Overall the results presented in this section in Table 2 through Table 8 offer robust evidence that an increase in the level of political competition is associated with an increase in the generosity of municipal pensions with no effects on the generosity of plans run by municipal authorities. In addition to the robustness checks reported here, the results are also robust to numerous checks not included in the paper but available on request, for example, to clustering standard errors at the municipality level or plan level or two-way clustering by county and year or two-way clustering by municipality and year. The results are also robust to splitting the sample based on the employee group covered – non-uniformed personnel versus policemen and firefighters.[51]

4.1.4. Alternative lag structures

Given that the data on pension benefits are not disaggregated by retiree and includes retirees from various cohorts, it is challenging to associate the average level of benefits received by retirees with the level of political competition for one specific year that corresponds to a unique lag structure. Based on Rauh (2017), I have used a lag structure of 11 years thus far but that is not the only defensible choice. Novy-Marx and Rauh (2011) suggests that the weighted average duration of pension liabilities is 13 years while Waring (2004a,b) suggests a longer duration of 15 years. Hence in this subsection, I present results with several different lag structures, which correspond to different assumptions about how quickly higher levels of political competition manifest itself in the form of more generous pension benefits. I use lags ranging from 0 to 3 years in Panels A through D of Table 9 (which implicitly assume a rapid process of adjustment) but also try out longer lags of 13 and 15 years in Panels E and F.

As we can see from the results in Table 9, the estimated effects of political competition are not particularly sensitive to the lag structure used. The coefficients on political competition continue to be statistically significant and of similar magnitude in panels A through E. The results do become weaker with a 15-year lag but that is perhaps to be expected given the nature of the underlying adjustment process by which political competition affects the generosity of pension benefits.

5. Examining the effects of political competition on defined contribution plans

Anecdotal evidence suggests that political influences are less influential in affecting the parameters for a DC plan compared to a DB plan.[52] For example, a report prepared in the context of reforming Florida's Retirement System (FRS) (Florida TaxWatch Report, 2013) notes:

> Another important benefit of the DC Investment Plan is that it is insulated from political temptations....Any benefit given under a DC plan must be paid for in that same year because it cannot be legally underfunded. This improves the financial health and security of the FRS because retirement assets belong to the individual state employees and are therefore not susceptible to the whims of the state.

In Table 10 therefore I examine the effects of political competition on the employer contribution rate for all DC plans from Pennsylvania for the period 2003–2013. I choose to focus on the employer contribution rate because for DC plans, it is not meaningful to talk of the average pension benefit received on retirement or the ratio of benefits to wages. The employer contribution rate to the DC plan is however a meaningful plan parameter as it reflects the extent to which an employer puts aside money each year and comes closest to our conception of generosity of a DC plan.

[50] Results are similar if I use the log of active members as weights or only analyze plans with 10 or more members and those are available on request.

[51] A deep-dive of the 213 municipal authority plans which appear in our dataset in the last year of our sample (2013) shows that about 70.0% are sanitary authorities or water authorities or authorities that provide both services simultaneously. The remaining plans mostly correspond to authorities that provide a wide array of miscellaneous services such as transportation, parking, or economic development. Police or fire services are not provided through municipal authorities (as the term is traditionally used) but I classify regional police departments (and the sole regional fire department) as authorities because they are somewhat insulated from the political influence that in my view distinguishes municipal authorities. Our results however are not sensitive to that choice. I find political competition does not have any effect on municipal authority plans, regardless of whether they cover non-uniformed personnel or policemen and firefighters.

[52] The decision of whether to offer a DB or a DC plan is, in itself, endogenous. In a set of regressions, using both OLS and probit estimation approaches, I find that an increase in political competition makes it more likely that a municipality offers a DB plan as compared to a DC plan. That result is consistent with the view that politicians in politically competitive jurisdictions desire to pass on the costs of pensions to future generations and the structure of DB (but not DC) plans makes that possible.

Table 9
Effect of political competition on log of average benefits using alternative lags.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Assuming no lag (political competition affects benefits contemporaneously)						
Political Competition	0.380***	0.526***	0.499***	−0.722	−0.209	−0.179
	(0.142)	(0.139)	(0.163)	(0.432)	(0.433)	(0.489)
Panel B: Assuming a lag of 1 year between political competition and benefit levels						
Political Competition	0.401**	0.564***	0.545***	−0.768	−0.158	−0.113
	(0.155)	(0.151)	(0.182)	(0.522)	(0.543)	(0.608)
Panel C: Assuming a lag of 2 years between political competition and benefit levels						
Political Competition	0.347**	0.460***	0.423***	−0.486	−0.0249	0.0222
	(0.134)	(0.132)	(0.153)	(0.464)	(0.464)	(0.486)
Panel D: Assuming a lag of 3 years between political competition and benefit levels						
Political Competition	0.369***	0.468***	0.434***	−0.565	−0.140	−0.0998
	(0.125)	(0.121)	(0.147)	(0.476)	(0.463)	(0.483)
Panel E: Assuming a lag of 13 years between political competition and benefit levels						
Political Competition	0.315***	0.227*	0.196*	0.159	0.143	0.187
	(0.118)	(0.123)	(0.114)	(0.249)	(0.252)	(0.283)
Panel F: Assuming a lag of 15 years between political competition and benefit levels						
Political Competition	0.236*	0.135	0.107	0.0711	0.0526	0.131
	(0.135)	(0.136)	(0.125)	(0.286)	(0.298)	(0.334)
Number of observations	8359	8359	8359	1141	1141	1141
Log of wages and employee contributions to pensions	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. County and year fixed effects, employee-group dummies, log of wages, and employee contributions to the pension plan (as a percentage of payroll) are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

I first estimate the effects of political competition on the employer contribution rate using data for all years and for all plans hewing exactly to specification (1).[53] Subsequent rows replicate the robustness checks that were conducted earlier on DB plans in Table 8 with each row corresponding to a different robustness check. Estimates for municipalities are presented in columns (1) through (3), with estimates for municipal authorities in columns (4) through (6). In the interest of brevity, I only present the coefficients on the variable representing the intensity of political competition and omit coefficients on the control variables. The complete results are available from the author on request.

As the coefficients on political competition indicate, DC plans appear less susceptible to political influence compared to DB plans. The coefficient on political competition for municipal DC plans is statistically insignificant in each of the 24 specifications presented in the table, in contrast to our previous set of findings on DB plans. This null result likely follows from the fact that with DC plans, it is hard for politicians to pass on the costs of a more generous plan onto future generations of taxpayers; a more generous DC plan requires a higher level of contributions today that have to be met from current tax revenues and politicians are less willing to make a DC plan more generous in order to avoid the risk of alienating voters.

[53] Given the fact that generosity of DC plans can change much more readily in response to changes in the political environment, my baseline results include contemporaneous measures of political competition, i.e. no lag. The results are however robust to varying lags as shown in Table A.3. In that table I present results with alternative lags, viz. 1–3 years in Panels A through C and longer lags of 11, 13, and 15 years in Panels D, E, and F respectively. Those results confirm the lack of a relationship between municipal-level political competition and the generosity of DC plans, regardless of the lag structure used.

Table 10
Effect of political competition on employer contribution rate of defined contribution (DC) plans.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Base Specification	−0.670	−0.984	−0.457	−1.941	0.102	0.393
	(1.322)	(1.366)	(1.216)	(2.562)	(2.392)	(2.036)
RC1: Controlling for average Democratic vote share	−0.598	−0.943	−0.396	0.293	0.299	0.391
	(1.375)	(1.402)	(1.203)	(2.578)	(2.289)	(2.121)
RC2: Using vote shares based solely on Presidential elections	0.0108	−0.233	0.829	−1.442	1.891	3.092
	(1.680)	(1.755)	(1.702)	(3.534)	(3.456)	(3.165)
RC3: Introducing county-by-year fixed effects	−0.398	−0.796	−0.112	−2.499	0.311	0.406
	(1.687)	(1.745)	(1.586)	(3.980)	(3.706)	(3.362)
RC4: Introducing municipality-specific linear time trends	−0.146	−0.216	−0.144	−0.0227	−0.0461	−0.0359
	(0.569)	(0.551)	(0.525)	(0.722)	(0.769)	(0.804)
RC5: Long differences (only including data for years) 2003 and 2013 with municipality fixed effects	−0.0636	−0.0838	−0.251	2.792	2.828	2.443
	(1.969)	(1.997)	(2.098)	(2.127)	(2.172)	(2.190)
RC6: Only including municipalities that have at least one authority	−1.469	−0.807	−1.109	−1.941	0.102	0.393
	(2.077)	(1.958)	(1.833)	(2.562)	(2.392)	(2.036)
RC7: Weighted least squares with log of active members as weights	−0.336	−0.627	−0.388	−1.995	0.570	0.674
	(1.164)	(1.120)	(1.066)	(2.594)	(2.395)	(1.979)
Log of wages	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, and log of number of active members		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined contribution pension plans from Pennsylvania for the period 2003–2013. The number of observations equals 3158 for the base specification, RC1, RC2, RC3, RC4, and RC7, 1062 for RC5, and 531 for RC6 for municipal plans (cols. (1)–(3)). The number of observations equals 1317 for the base specification, RC1, RC2, RC3, RC4, RC6 and RC7, and 442 for RC5 for plans run by municipal authorities (cols. (4)–(6)). The dependent variable is the employer contribution rate to the defined contribution plan, expressed as a percentage of payroll. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has not been lagged. County and year fixed effects, employee-group dummies, and log of wages are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, and the log of number of active members. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

Political competition also has no effect on the employer contribution rate of DC plans run by municipal authorities given that the insulation of authority boards from political influence is now coupled with the transparency of DC plans.

6. A sample-split test examining differential effects based on voter awareness

Models of fiscal illusion rely on the existence of informational frictions. If individuals had perfect information about politicians' platforms and were rational, then the political budget cycles modeled by Rogoff (1990) or Rogoff and Sibert (1988) would simply not exist. The model in Glaeser and Ponzetto (2014), which this paper most directly builds on, also hinges on information asymmetries. The authors argue that pension obligations are shrouded because of lower availability of information about pensions than wages and because voters find it challenging to understand the impact of unfunded pension obligations on their future tax burdens. This insight forms the basis of an empirical test. By splitting the sample of municipalities into two groups based on the level of voter awareness and information, we can examine if the effects of political competition are larger in places that have a higher proportion of uninformed voters. To operationalize this test, I look for variables that likely reflect variations among the residents of a municipality in their ability to understand the nuances of local public finance.

A literature review suggests that newspapers have been historically the most important source of information about state and local politics and that this has been true even in the television era (Gentzkow et al., 2011). As Hayes and Lawless (2015) note: "When local news outlets like daily newspapers devote less coverage, and less substantive coverage, to politics—whether it be about a House race, state legislative contest, or municipal election—there are few alternative sources to which citizens can turn....In the vast majority of U.S. communities, there is no local Politico, no local Talking Points Memo, no local Hot Air." In stressing the role of local print media, they note that "unlike with national elections, there are virtually no other widely available outlets to which people can turn for information about local politics."

Accordingly, based on this review of the literature, I turn to newspaper penetration as a variable which can help us distinguish places that have a large number of well-informed voters from places with relatively few well-informed voters. Using data on newspaper circulation that were generously provided by the Alliance for Audited Media, I am able to obtain estimates of newspaper penetration at the municipal level and I use that variable to split the sample of plans into two

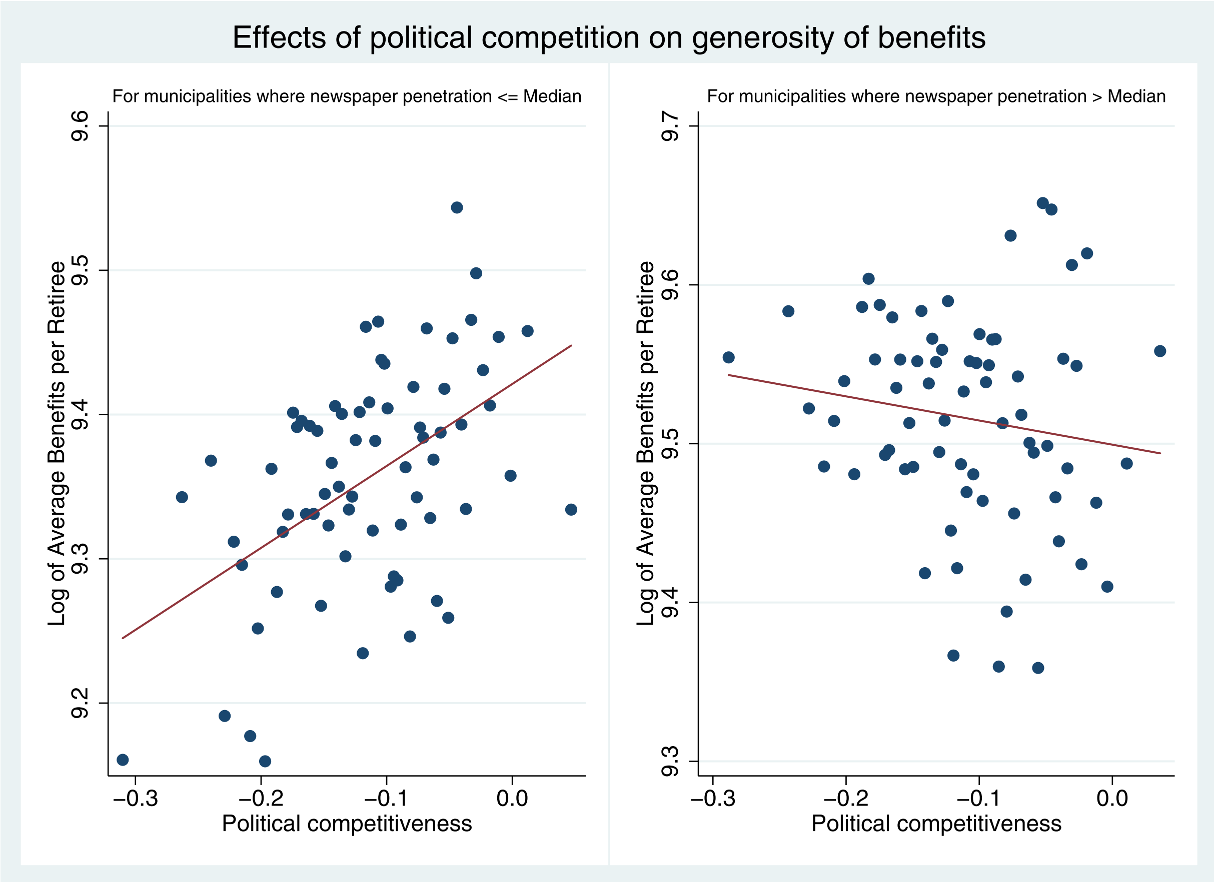

Fig. 3. Effects of political competition on benefit levels for municipal plans that differ by voter awareness. These plots were drawn using the Stata command, binscatter, with 70 bins used for both sub-samples of municipal plans. They correspond to the regression results presented in Columns (3) and (6) of Panel A of Table 11 in which I regress log average benefit on political competition, controlling for log wages, member contributions to the plan, employee-group dummies, plan-level controls (a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries), and socioeconomic controls (the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation), along with dummy variables for the class of municipality, and county and year fixed effects.

groups.[54] The first group includes plans for which their municipality's newspaper penetration is less than (or equal) to the median and a second group that includes plans for which newspaper penetration is higher than the median. In light of the discussion above, I characterize the first group as having a higher proportion of less informed voters and the second group as having a higher proportion of more informed voters.

Table 11 presents the results obtained by splitting municipal plans in the manner described above with Panels A, B, C, and D providing results for log benefits, the ratio of benefits to wages, the unadjusted "as-is" normal costs, and the measure of lifetime value of pension benefits (constructed per (2)) respectively. In all panels, columns (1) through (3) present the coefficient on political competition for plans where newspaper penetration is less than or equal to the median whereas columns (4) through (6) present the coefficient for plans where newspaper penetration is more than the median. The last three columns of the table examine if coefficients in columns (1)–(3) are statistically different from the corresponding coefficients in columns (4)–(6).

The results in Table 11 suggest that the effects of political competition are larger in municipalities where newspaper penetration is less than or equal to the median, whereas they are muted (in fact, absent) in municipalities where newspaper penetration is higher than the median, with the differences in coefficients statistically significant in Panels A and B.[55] Importantly these results do not hinge on places with lower newspaper penetration having more generous plans on average;

[54] These data are at the zipcode level and are based on audits of circulations conducted between June 2010 and June 2012. To obtain estimates of newspaper penetration at the municipal level, I use a Census-created crosswalk from zipcode to municipality.

[55] I also get similar results if I estimate a regression on the entire sample by interacting the level of political competition with newspaper penetration. Political competition is positive and statistically significant but the interacted term is negative and statistically significant, with a magnitude similar to that

Table 11
Sample-split test examining the effects of political competition on plan generosity.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Newspaper penetration <= Median			Newspaper penetration > Median			p-value coefficients differ across samples		
Panel A: Effect of Political Competition on Log of Average Benefits									
Political Competition	0.678***	0.603***	0.568***	−0.0171	−0.142	−0.152	0.0036	0.0008	0.0012
	(0.196)	(0.187)	(0.183)	(0.155)	(0.147)	(0.146)			
Panel B: Effect of Political Competition on Ratio of Benefits to Wages									
Political Competition	0.154***	0.135***	0.136***	−0.0235	−0.0296	−0.0300	0.0032	0.0049	0.0068
	(0.0467)	(0.0426)	(0.0414)	(0.0452)	(0.0452)	(0.0457)			
Number of observations for panels A and B	4213	4213	4213	4122	4122	4122			
Panel C: Effect of Political Competition on Normal costs (unadjusted, "as-is")									
Political Competition	2.430*	2.524*	2.430**	0.606	0.544	0.215	0.2995	0.2272	0.1528
	(1.300)	(1.276)	(1.199)	(1.431)	(1.309)	(1.204)			
Panel D: Effect of Political Competition on Normal costs adjusted for variations in interest rate assumptions and member contributions to the plan									
Political Competition	5.335*	5.289*	4.901*	2.872	0.780	0.129	0.5284	0.1780	0.1296
	(2.678)	(2.722)	(2.598)	(3.317)	(2.470)	(2.304)			
Number of observations for panels C and D	5601	5601	5601	5067	5067	5067			
County and year FE	✔	✔	✔	✔	✔	✔			
Employee-group dummies	✔	✔	✔	✔	✔	✔			
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, and log of number of active members and beneficiaries		✔	✔		✔	✔			
Municipal demographic and fiscal controls			✔			✔			

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
The p-value in columns (7) - (9) is generated from statistical tests which examine whether the coefficient on political competition in columns (1) through (3) differs from corresponding coefficients in columns (4) through (6). Regressions estimated on all municipal defined benefit pension plans from Pennsylvania for the period 2003–2013. The dependent variable in Panels A and B are the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries) and the ratio of the average benefits received by retirees to the wages received by active members. The dependent variable in Panels C and D are normal costs (as-is) and normal costs, adjusted for variations in interest rate assumptions and member contributions to the plan as noted in the text. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. The sample splits are based on newspaper penetration constructed using circulation counts provided by the Alliance for Audited Media. These data are at the zipcode level and are based on audits of circulations conducted between June 2010 and June 2012. County and year fixed effects and employee-group dummies are included in all specifications in all panels. Log of wages is controlled for in Panels A, C, and D while employee contributions to the pension plan (as a percentage of payroll) is controlled for in Panels A and B. Plan-specific controls included in columns (2), (3), (5), and (6) of all panels are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) of all panels are the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

those places in fact have *less* generous plans on average. What these regressions estimate is the *strength of the relationship* between political competition and plan generosity *within each sub-sample*. Given that our findings hold up in the most complete specification in which I control for a variety of municipal characteristics such as per capita income, homeownership rates, and income inequality, one cannot simply explain away these sample-split results as reflecting differences in socioeconomic characteristics between municipalities that have high newspaper penetration from those with relatively low newspaper penetration rates.[56] Furthermore, when I replicate these steps for plans run by municipal authorities, I observe that across both sub-samples, political competition has no effect on pension benefits. That result is consistent with the claim that the differential effects of political competition on municipal plans are being driven by variation in the level of information available to voters rather than by differences in socioeconomic characteristics of the municipalities themselves.[57]

Given that newspaper readership is likely to correlate with voter awareness and engagement, these results suggest that when faced with an electorate that is more aware and informed of the true cost of unfunded pension obligations, politi-

on the uninteracted term. The inference from such a regression is that political competition is associated with an increase in the generosity of benefits only for places where newspaper penetration is below the median; there are no effects in places where newspaper penetration is above the median.

[56] I additionally examine the sensitivity of the results to including measures of geographic mobility and educational attainment but find that controlling for them does not make any meaningful difference to any of our estimates.

[57] Those additional results for municipal authority sample splits are available from the author on request.

cians are less likely to underfund public pensions. This finding is consistent with the key assumption in Glaeser and Ponzetto (2014) as it pertains to limited information on the part of voters and provides empirical support for that assumption. The results from Panel A of Table 11 can be captured in the form of Fig. 3, which illustrates that the effects of political competition emerge only in places that have a higher proportion of relatively uninformed voters.

7. Conclusions

This paper proposes that political competition plays a key role in influencing the fiscal health of defined benefit pensions offered to public-sector employees. In their desire to win re-election, politicians in politically competitive jurisdictions promise generous benefits to public-sector workers while failing to make the contributions necessary to fund them fully in order to avoid having to raise taxes. Using a panel dataset for 3000 local plans from Pennsylvania for the period 2003–2013, I obtain evidence that is consistent with this hypothesis. I find that more politically competitive municipalities have more generous benefits, with a one standard deviation increase in the level of political competition associated with an increase of about 3 percent in benefits (about $340–504 per retiree per year). These results hold up to controlling for variations across plans in their degree of coverage under collective bargaining and plan size and across municipalities in their levels of economic prosperity, age structure, income inequality, and ethnic fragmentation, making it unlikely that heterogeneity across municipalities or plans are driving these results. These results are also robust to a host of checks such as defining pension generosity differently or using alternative measures of political competition or to the inclusion of municipality fixed effects or county-by-year fixed effects or municipality-specific linear time trends.

In contrast to the robust effects of political competition on the generosity of municipal DB plans, I fail to discern any relationship between the level of political competition of municipal authorities and the generosity of the plans that they offer. This result is likely driven by the fact that members of municipal authority boards are appointed and hence far removed from the electoral will of voters, at least when compared to mayors and council members who make decisions on municipal pensions. The sharply different effects of political competition for municipal authorities when compared with municipalities are in line with papers that have examined differences between appointed and elected officials. These include Besley and Coate (2003) who find that public utility regulators who are elected are more pro-consumer in their regulatory policies, Lim (2013) who finds that the harshness of sentences awarded by elected judges is strongly related to the political ideology of the voters in their districts, while that of appointed judges is not, and Dove et al. (2018) who find that having more elected members on a pension system' s board of trustees is consistently associated with lower bond ratings (and higher borrowing costs) compared to having appointed and ex-officio board members.[58] Indeed, the idea that holding officials responsible for complex policy areas directly accountable to voters through election can result in lower levels of performance as suggested by Whalley (2013) appears relevant here and perhaps introducing a degree of political insulation between voters and officials responsible for making decisions on public pensions is an idea worth pursuing.

The desirability of minimizing the influence of electoral politics on the management of public pension plans is buttressed by the null results of political competition on the generosity of defined contribution plans. Indeed, because a more generous DC plan requires higher contributions today rather than several years in the future, the structure of such plans constrains the ability of politicians to pass on the costs of current labor services to future taxpayers. Thus, moving from DB plans to DC plans that I find as less susceptible to political influence may be a useful step that preserves inter-generational equity and reduces the likelihood that future taxpayers will experience tax increases or service cuts to make up for shortfalls in the pension fund (Novy-Marx and Rauh, 2014). Such motivations appear to have been at work for Utah which ended its traditional DB plan for new workers and offered them a choice between a DC plan and a hybrid DB-DC plan or Rhode Island which created a hybrid DB-DC plan with a lower guaranteed pension supplemented by a DC component (McGuinn, 2014). Similar reforms continue to be debated and discussed across state and local governments throughout the country reflecting the rapidly growing costs they face for public pensions.

Given the salutary effects of competition in other contexts, it is an interesting question why increased political competition does not result in shining a brighter light on the fiscal practices of incumbents and moderate the growth rate of benefits. Possible explanations rest in part on voter apathy but also on the availability of information in a format that is easy to comprehend and act upon. In a world in which voters find it challenging to understand the complexity of public pensions and the media and credit rating agencies do not accord public pensions the importance they deserve, political competition fails to have its usual beneficial effects.[59] The increased attention that has been given to public pensions of late following the Great Recession and municipal bankruptcies, Detroit's in particular, may result in a different dynamic in the future but by and large, for the sample period analyzed in this paper, increased political competition did not result in lower benefits and all our results suggest very much the opposite.

---

58 Andonov et al. (2018) also examine pension boards and find that state pension boards of trustees that are heavily populated by politicians underperform in terms of their private equity investments in part because of the incentives faced by these politicians.

59 For example, it wasn't until April 2013 that Moody's "announced its final approach to the way it will analyze and adjust pension liabilities as part of its credit analysis of state and local governments" reflecting their view that "pension obligations are a significant source of credit pressure for governments and warrant a more conservative view of the potential size of the obligations." https://www.moodys.com/research/Moodys-announces-new-approach-to-analyzing-state-local-government-pensions--PR_271186.

Table A.1
Effect of political competition on log of average benefits and normal costs for DB plans run by municipal authorities, split based on the nature of the match with municipalities.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for plans which do not have an unambiguous municipal match			Coefficient for plans which have an unambiguous match with one municipality		
Panel A: Dependent variable: Log of average benefits						
Political Competition	0.0104	0.00433	−0.198	0.347	0.271	0.326
	(0.454)	(0.469)	(0.300)	(0.417)	(0.424)	(0.384)
Number of observations	582	582	582	559	559	559
R^2	0.62	0.64	0.66	0.44	0.49	0.54
Panel B: Dependent variable: Normal costs, unadjusted (as-is)						
Political Competition	−1.041	−0.717	−2.036	0.937	0.0681	−0.752
	(2.618)	(2.102)	(1.722)	(2.726)	(2.846)	(2.435)
Number of observations	767	767	767	754	754	754
R^2	0.72	0.76	0.78	0.32	0.36	0.39
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania run by municipal authorities for the period 2003–2013. The dependent variable in Panel A is the log of the average benefits received by retirees, while in Panel B, the dependent variable is the unadjusted normal costs. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. County and year fixed effects, employee-group dummies, and log of wages are included in all specifications, with employee contributions to the pension plan (as a percentage of payroll) additionally included in Panel A. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries. Municipal demographic and fiscal controls included in columns (3) and (6) are the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

Beyond these policy implications, the paper contributes to the literature of public-sector labor markets by highlighting the role played by political competition in generating compensation structures that are backloaded. However, as our sample-split analysis using newspaper penetration data suggests, voter awareness and engagement can moderate the negative effects associated with political competition and local newspapers can play a crucial role in generating such awareness and engagement. The demise of local newspapers which has been noted by many observers (e.g. Schulhofer-Wohl and Garrido, 2013) is therefore of concern for those who care about the health of local democracy.

Appendix A. Using a matching-type estimator to estimate the effects of political competition on the generosity of pension plans

I lay out the steps followed in constructing an estimate of the effects of political competition inspired by the literature on matching.[60] This empirical approach is geared to address concerns that there are observed and unobserved differences between politically competitive and uncompetitive municipalities which make it hard to attribute causality to our estimates. In most applications where matching is used we have a binary treatment variable that distinguishes treated and control units. That is different here given that political competition, our key independent variable of interest, is continuous rather than discrete. I can however use a cut-off value of political competition and use that to split the sample of municipal plans into two groups: one with a "low" level of political competition and another with a "high" level of political competition and then examine if benefits are less generous in the group that has a low level of competition. Where the techniques of matching come in are that rather than comparing these groups as-is, we can construct a matched pair by identifying two plans which look "similar" on underlying municipal characteristics, and yet differ in competitiveness such that plan 1 belongs to group 1 with levels of political competition lower than the cut-off and its matched pair belongs to group 2 with levels of political competition higher than the cut-off. More specifically, these are the steps I undertake:

1. Compute the average level of political competition for any municipality that offers a DB municipal pension plan at any point during the sample period (2003–2013).

[60] Matching has been used extensively in evaluation of the effectiveness of active labor market programs (e.g. LaLonde, 1986; Heckman and Hotz, 1989; Dehejia and Wahba, 1999, 2002 and Smith and Todd, 2005).

Table A.2
Effect of political competition on municipal contributions for DC plans run by municipal authorities, split based on the nature of the match with municipalities.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for plans which do not have an unambiguous municipal match			Coefficient for plans which have an unambiguous match with one municipality		
Dependent variable: Employer Contribution Rate to the pension plan (as a percentage of payroll)						
Political Competition	−1.280	−0.785	−1.091	−4.508	−1.568	0.144
	(2.961)	(3.095)	(2.254)	(4.421)	(4.408)	(3.400)
Number of observations	821	821	821	496	496	496
R^2	0.39	0.51	0.53	0.44	0.48	0.58
Log of wages	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, and log of number of active members		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined contribution pension plans from Pennsylvania run by municipal authorities for the period 2003–2013. The dependent variable is the employer contribution rate to the defined contribution plan, expressed as a percentage of payroll. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has not been lagged. County and year fixed effects, employee-group dummies, and log of wages are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, and the log of number of active members. Municipal demographic and fiscal controls included in columns (3) and (6) are the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

2. Compute the median of this distribution and split plans into two groups using this value. Group 1 includes plans for which the average level of political competition is less than (or equal to) the median (and is characterized as having "low" levels of political competition) while group 2 includes the remaining plans (and is characterized as having "high" levels of political competition).
3. Construct standardized normal variables for the socioeconomic characteristics used in the regressions: the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, revenues per capita, the level of income inequality, and the level of ethnic fragmentation. Doing so is essential for us to compute Euclidean-distances between any two plans using these characteristics.
4. Calculate distances between all possible pairs of plans in groups 1 and 2 taking care to only match plans that pertain to the same year and the same employee group. The distance measure used is the sum of squares of differences in the values of these characteristics between a plan in group 1 and that in group 2. Retain the match with the lowest distance and discard all remaining matches. This is akin to nearest neighbor matching with replacement.
5. Construct variables which measure the difference in the level of benefits and the level of political competition between the matched plans in the two groups. Regress the difference in the level of benefits on the difference in the level of political competition, controlling for year and employee-group fixed effects, the level of wages in the two municipalities, and member contributions to the plans. Augment the regression progressively with more controls, such as the socioeconomic characteristics of the municipalities to which these plans belong.
6. Replicate the same set of steps laid out for municipal authorities in order to examine the effects of political competition on the generosity of plans run by municipal authorities in this matching-inspired framework.

Table A.4 presents the results we obtain following the steps described above. Columns (1) through (3) present the results for municipal plans, with columns (4) through (6) presenting the results for municipal authorities. I do not use regression weights in Panel A, while in Panel B the inverse of the Euclidean-distance between the matched plans serves as a weight.

As we can see, the estimated effects of political competition on municipal plans are positive and statistically significant at conventional levels of significance across all specifications. In the unweighted regressions, the coefficients on political competition are clustered in a range from 0.381 to 0.511, similar to the range of 0.282–0.418 I report in the baseline results in Table 2. Considering a one standard deviation change in the independent variable – the difference in the level of political competition between a plan in group 1 and its matched pair in group 2 – the estimated effect of an increase in political competition lies between 3.2 and 4.3 percent.[61] When we use weighted regressions, the coefficients are larger in magnitude ranging from 0.454 to 0.648 and the estimated effect of an increase in political competition by one standard deviation averages 4.4 percent.

[61] For example, using the coefficient in col. (1) of 0.511 and multiplying that with 0.0838 corresponding to the standard deviation of the difference in political competition between matched plans, we get 4.28%.

Table A.3
Effect of political competition on employer contribution rate of defined contribution (DC) plans using alternative lags.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Assuming a lag of 1 year between political competition and employer contribution rate						
Political Competition	−0.512	−0.910	−0.343	−2.260	0.0607	0.147
	(1.552)	(1.605)	(1.486)	(3.264)	(2.987)	(2.779)
Panel B: Assuming a lag of 2 years between political competition and employer contribution rate						
Political Competition	−0.689	−0.965	−0.554	−1.174	0.315	0.265
	(1.273)	(1.326)	(1.228)	(2.543)	(2.344)	(2.115)
Panel C: Assuming a lag of 3 years between political competition and employer contribution rate						
Political Competition	−1.040	−1.383	−1.038	−0.688	0.184	0.265
	(1.384)	(1.441)	(1.373)	(2.471)	(2.223)	(1.976)
Panel D: Assuming a lag of 11 years between political competition and employer contribution rate						
Political Competition	−1.023	−1.013	−0.967	−1.542	−2.765	−3.392*
	(0.954)	(1.001)	(1.000)	(1.898)	(1.775)	(1.843)
Panel E: Assuming a lag of 13 years between political competition and employer contribution rate						
Political Competition	−0.855	−0.863	−1.001	−1.502	−2.184	−2.772*
	(0.918)	(0.955)	(0.956)	(1.758)	(1.550)	(1.575)
Panel F: Assuming a lag of 15 years between political competition and employer contribution rate						
Political Competition	−1.395	−1.447	−1.624*	−0.952	−1.782	−2.208
	(0.972)	(1.009)	(0.970)	(1.852)	(1.542)	(1.589)
Log of wages	✔	✔	✔	✔	✔	✔
County and year FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Social Security coverage, coverage under collective bargaining, and log of number of active members		✔	✔		✔	✔
Municipal demographic and fiscal controls			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined contribution pension plans from Pennsylvania for the period 2003–2013. The dependent variable is the employer contribution rate to the defined contribution plan, expressed as a percentage of payroll. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. County and year fixed effects, employee-group dummies, and log of wages are included in all specifications. Plan-specific controls included in columns (2), (3), (5), and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, and the log of number of active members. Municipal demographic and fiscal controls included in columns (3) and (6) are the percentage of households that are owner-occupied, the log of per capita income, the percentage of population aged 65 or older, the unemployment rate, the fraction of tax revenues spent on debt service, the log of revenues per capita, the level of income inequality, and the level of ethnic fragmentation, along with dummy variables for the class of municipality. The dependent variables and all control variables have been winsorized at the 2.5% and 97.5% levels.

The identical exercise when conducted for plans run by municipal authorities confirms that political competition does not have any effect on the generosity of such plans with coefficients far from statistical significance. Thus, on the basis of the results presented in Table A.4, one can be more confident in the conclusion that an increase in political competition is associated with an increase in the generosity of municipal defined benefit plans, but not for plans run by municipal authorities.

References

Alesina, A., Baqir, R., Easterly, W., 2000. Redistributive public employment. J. Urban Econ. 48 (2), 219–241.
Andonov, A., Hochberg, Y.V., Rauh, J.D., 2018. Political representation and governance from the investment decisions of public pension funds. J. Finance 73 (5), 2041–2086.
Anzia, S.F., Moe, T.M., 2015. Public sector unions and the costs of government. J. Politics 77 (1), 114–127.
Anzia, S.F., Moe, T.M., 2017. Polarization and policy: the politics of public-sector pensions. Legislative Stud. Q. 42 (1), 33–62.
Bagchi, S., 2017. The Effects of Political Competition on the Funding of Public-Sector Pension Plans. Villanova School of Business Department of Economics and Statistics, Villanova, PA. Working Paper No. 36.
Banzhaf, H.S., Oates, W.E., 2013. On fiscal illusion in local public finance: re-examining Ricardian equivalence and the renter effect. Natl. Tax J. 66 (3), 511–540.
Bennett, J.T., Dilorenzo, T.J., 1982. Off-budget activities of local government: the bane of the tax revolt. Public Choice 39 (3), 333–342.
Bertrand, M., Duflo, E., Mullainathan, S., 2004. How much should we trust differences-in-differences estimates? Q. J. Econ. 119 (1), 249–275.
Besley, T., Coate, S., 2003. Elected versus appointed regulators: theory and evidence. J. Eur. Econ. Assoc. 1 (5), 1176–1206.
Besley, T., Persson, T., Sturm, D.M., 2010. Political competition, policy and growth: theory and evidence from the US. Rev. Econ. Stud. 77 (4), 1329–1352.
Bewerunge, P., Rosen, H.S., 2013. Wages, pensions, and public-private sector compensation differentials for older workers. Public Admin. Res. 2 (2), 233–249.

Table A.4
Effects of political competition on plan benefits for municipalities and for municipal authorities using matching-type estimator.

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Unweighted regressions						
Political Competition	0.511** (0.199)	0.381* (0.201)	0.381* (0.193)	0.511 (0.688)	0.562 (0.701)	0.297 (0.601)
Panel B: Weighted regressions where weight = inverse of Euclidean-distance between matched pairs						
Political Competition	0.648*** (0.201)	0.454** (0.207)	0.475** (0.198)	0.248 (0.526)	0.460 (0.502)	0.238 (0.529)
Number of observations	4182	4182	4182	750	750	750
County and Year fixed effects	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Wages and employee contributions of matched plans		✔	✔		✔	✔
Social Security coverage, coverage under collective bargaining, actuarial funded ratio, log of number of active members, and log of number of beneficiaries for matched plans		✔	✔		✔	✔
Municipal demographic and fiscal controls of matched municipalities			✔			✔

Robust standard errors are clustered at the county level and are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all defined benefit pension plans from Pennsylvania for the period 2003–2013. Columns (1)–(3) pertain to DB plans offered by municipalities and those in columns (4)–(6) pertain to DB plans offered by municipal authorities. The dependent variable is the log of the average benefit received by all retirees (including Deferred Retirement Option Plan (DROP) beneficiaries but excluding disability, surviving spousal, and surviving child beneficiaries). The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$ and has been lagged by 11 years. County and year fixed effects, and employee-group dummies are included in all specifications, along with the log of wages and member contributions for both plans – those in Group 1 (with "low" political competition) and its matched pair in Group 2 (with "high" political competition). Plan-specific controls included in columns (2), (3), (5) and (6) are a dummy variable for Social Security coverage, the fraction of employees organized under collective bargaining, the actuarial funded ratio, the log of number of active members, and the log of number of beneficiaries for both plans. Municipal demographic and fiscal controls included in columns (3) and (6) are the log of per capita income, the fraction of tax revenues spent on debt service, the unemployment rate, the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the log of revenues per capita, the level of income inequality, the level of ethnic fragmentation, and the class of municipality for both municipalities. The dependent variable and all control variables have been winsorized at the 2.5% and 97.5% levels.

Bureau of Labor Statistics (BLS), 2010. Program perspectives on defined contribution plans. Program Perspect. 2 (6), 1–4.
Boyne, G.A., 1998. Party competition and local spending decisions. Br. J. Political Sci. 28 (1), 210–222.
Buchanan, J.M., 1960. Fiscal Theory and Political Economy. University of North Carolina Press, Chapel Hill, NC.
Buchanan, J.M., 1967. Public Finance in Democratic Process. University of North Carolina Press, Chapel Hill, NC.
Congleton, R., 2001. Rational ignorance, rational voter expectations, and public policy: a discrete informational foundation for fiscal illusion. Public Choice 107 (1–2), 35–64.
Congleton, R.D., Shughart II, W.F., 1990. The growth of social security: electoral push or political pull? Econ. Inq. 28 (1), 109–132.
Dehejia, R.H., Wahba, S., 1999. Causal effects in nonexperimental studies: reevaluating the evaluation of training programs. J. Am. Stat. Assoc. 94 (448), 1053–1062.
Dehejia, R.H., Wahba, S., 2002. Propensity score-matching methods for nonexperimental causal studies. Rev. Econ. Stat. 84 (1), 151–161.
Department of Community and Economic Development, 2015. Municipal Authorities in Pennsylvania. Governor's Center for Local Government Services, Harrisburg, PA.
Diebold, J., Reitano, V., McDonald, B., 2018. Sweat the small stuff: strategic selection of pension policies used to defer required contributions. Contemp. Econ. Policy 36 (3), 505–525. Special Issue on Aging
Dove, J., Collins, C., Smith, D.J., 2018. The impact of public pension board of trustee composition on state bond ratings. Econ. Governance 19 (1), 51–73.
Ehrenberg, R.G., Schwarz, J.L., 1986. Public-sector labor markets. In: Ashenfelter, O.C., Layard, R. (Eds.), Handbook of Labor Economics, vol. 2. Elsevier, Philadelphia, pp. 1219–1260. Chapter 22
Epple, D., Schipper, K., 1981. Municipal pension funding: a theory and some evidence. Public Choice 37 (1), 141–178.
Fischel, W.A., 2001. The Homevoter Hypothesis. Harvard University Press, Cambridge, MA.
Fitzpatrick, M.D., 2017. Pension-spiking, free-riding, and the effects of pension reform on teachers' earnings. J. Public Econ. 148 (C), 57–74.
Frandsen, B.R., 2016. The effects of collective bargaining rights on public employee compensation: evidence from teachers, firefighters, and police. ILR Rev. 69 (1), 84–112.
Gentzkow, M., Shapiro, J.M., Sinkinson, M., 2011. The effect of newspaper entry and exit on electoral politics. Am. Econ. Rev. 101 (7), 2980–3018.
Gittleman, M., Pierce, B., 2012. Compensation for state and local government workers. J. Econ. Perspect. 26 (1), 217–242.
Glaeser, E.L., Ponzetto, G.A.M., 2014. Shrouded costs of government: the political economy of state and local public pensions. J. Public Econ. 116 (C), 89–105.
Gustman, A.L., Steinmeier, T.L., 1993. Pension portability and labor mobility: evidence from the survey of income and program participation. J. Public Econ. 50 (3), 299–323.
Gustman, A.L., Steinmeier, T.L., 1995. Pension Incentives and Job Mobility. Kalamazoo, MI: W.E. Upjohn Institute for Employment Research.
Hayes, D., Lawless, J.L., 2015. As local news goes, so goes citizen engagement: media, knowledge, and participation in US house elections. J. Politics 77 (2), 447–462.

Heckman, J.J., Hotz, V.J., 1989. Choosing among alternative nonexperimental methods for estimating the impact of social programs: the case of manpower training. J. Am. Stat. Assoc. 84 (408), 862–874.
Johnson, R.N., Libecap, G.D., 1991. Public sector employee voter participation and salaries. Public Choice 68 (1/3), 137–150.
Kelley, D., 2014. The political economy of unfunded public pension liabilities. Public Choice 158 (1/2), 21–38.
LaLonde, R.J., 1986. Evaluating the econometric evaluations of training programs with experimental data. Am. Econ. Rev. 76 (4), 604–620.
Lazear, E.P., 1990. Pensions and deferred benefits as strategic compensation. Ind. Relat. 29 (2), 263–281.
Lazear, E.P., Moore, R.L., 1988. Pensions and turnover. In: Bodie, Z., Shoven, J.B., Wise, D.A. (Eds.), Pensions in the U.S. Economy. University of Chicago Press, Chicago, pp. 163–190. Chapter 6
Lewis, H.G., 1990. Union/nonunion wage gaps in the public sector. J. Labor Econ. 8 (1), S260–S328. Part 2: Essays in Honor of Albert Rees:
Lim, S.H., 2013. Preferences and incentives of appointed and elected public officials: evidence from state trial court judges. Am. Econ. Rev. 103 (4), 1360–1397.
Lovenheim, M.F., 2009. The effect of teachers unions on education production: evidence from union election certifications in three midwestern states. J. Labor Econ. 27 (4), 525–587.
McGuinn, P., 2014. Pension Politics: Public Employee Retirement System Reform in Four States. Brown Center on Education Policy. Working Paper. Brookings Institution, Washington DC
Meredith, M., Grissom, J., 2010. Partisanship in Local Elections: Regression Discontinuity Estimates from Unconventional School Board Races. Working Paper
Mulligan, C.B., Sala-i Martin, X., 1999. Gerontocracy, Retirement, and Social Security. NBER. Working Paper No. 7117. Cambridge, MA
Munnell, A.H., Aubry, J.-P., Hurwitz, J., Quinby, L., 2011. Unions and Public Pension Benefits. CRR. Working Paper No. 19. Center for Retirement Research, Boston College, Chestnut Hill, MA
Munnell, A.H., Haverstick, K., Soto, M., 2007. Why Have Defined Benefit Plans Survived in the Public Sector. CRR. Working Paper No. 2. Center for Retirement Research, Boston College, Chestnut Hill, MA
Nordhaus, D.W., 1975. The political business cycle. Rev. Econ. Stud. 42 (2), 169–190.
Novy-Marx, R., Rauh, J., 2011. Public pension promises: how big are they and what are they worth? J. Finance 66 (4), 1211–1249.
Novy-Marx, R., Rauh, J., 2014. Linking benefits to investment performance in US public pension systems. J. Public Econ. 116, 46–61.
Nuschler, D., Shelton, A.M., Topoleski, J.J., 2011. Social security: Mandatory coverage of new state and local government employees. CRS Report R41936. Congressional Research Service, Washington, DC.
Oates, W.E., 1988. On the nature and measurement of fiscal illusion. In: Brennan, G., Grewal, B.S., Groenewegen, P.D. (Eds.), Taxation and Fiscal Federalism: Essays in Honour of Russell Mathews. Australian National University Press, Sydney, Australia, pp. 64–82.
Oliver, E.J., Shang, E.H., 2007. Vote choice in suburban elections. Am. Political Sci. Rev. 101 (3), 393–408.
Public Plans Data, 2001–2016. Center for Retirement Research at Boston College, Center for State and Local Government Excellence, and National Association of State Retirement Administrators
Puviani, A., 1903. Teoria della Illusione Finanziaria [Theory of The Financial Illusion]. ISEDI Istituto Editoriale Internazionale, Milan, Italy.
Rauh, J.D., 2017. Hidden Debt, Hidden Deficits: How Pension Promises are Consuming State and Local Budgets. Hoover Institution Essay. Hoover Institution, Stanford, CA.
Florida TaxWatch Report, 2013. Modernizing the Florida Retirement System: Switching to a Defined Contribution Plan. Tallahassee, FL
Rogoff, K., 1990. Equilibrium political budget cycles. Am. Econ. Rev. 80 (1), 21–36.
Rogoff, K., Sibert, A., 1988. Elections and macroeconomic policy cycles. Rev. Econ. Stud. 55 (1), 1–16.
Sabin, M.J., 2015. Backtested pension math: an empirical look at the causes of CalPERS underfunding. J. Retirement 2 (3), 40–54.
Schulhofer-Wohl, S., Garrido, M., 2013. Do newspapers matter? short-run and long-run evidence from the closure of The Cincinnati Post. J. Media Econ. 26 (2), 60–81.
Sieg, H., Wang, Y., 2013. The impact of unions on municipal elections and urban fiscal policies. J. Monetary Econ. 60 (5), 554–567.
Smith, J.A., Todd, P.E., 2005. Does matching overcome Lalonde's critique of nonexperimental estimators? J. Econom. 125 (1–2), 305–353.
Solon, G., Haider, S.J., Wooldridge, J.M., 2015. What are we weighting for? Journal of Human Resources 50 (2), 301–316.
Waring, M.B., 2004a. Liability-relative investing. J. Portfolio Manage. 30 (4), 8–20.
Waring, M.B., 2004b. Liability-relative investing II. J. Portfolio Manage. 31 (1), 40–53.
Whalley, A., 2013. Elected versus appointed policy makers: evidence from city treasurers. J. Law Econ. 56 (1), 39–81.